Change Log

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Corrections, release repairs and documentation changes

Draft 1.6, release 2026-09-19 · document revision 2026-09-19

`release 2026-09-06 · 51 differential extractions + 6 one-bin totals · blind`

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Abstract

This document records every correction that materially moved a number in the CC$\pi^{\pm}$ NuMI analysis, with the value before and after, together with the release repairs and documentation changes made in the same period; entries of the latter kind are labelled as such in their headings ("figure only", "presentation only", "documentation-only"). It exists so that a reader of the analysis note never has to work out whether a given table predates a given fix, and so that a result quoted from an older draft can be identified as superseded rather than merely different.

Most entries are normalisation errors. The extraction divides by four numbers (the data exposure, the integrated flux, the argon target count and the per-run Monte Carlo exposure), and eight separate defects came from one of those four differing between configurations without anything detecting it. The response is `norm_manifest.py`, which reads the authoritative configuration files and asserts that they agree, so that a future drift fails a check instead of producing a wrong cross section.

Three corrections changed physics conclusions rather than only normalisation, and the conclusions they invalidated are marked as withdrawn here and in the analysis note: the non-uniform-grid regularisation defect, which had collapsed several additional-smearing matrices towards rank one; the beam-axis convention, under which reconstructed angular and transverse observables were not the same quantities as the generator predictions they were being compared against; and the double application of A_C to the model predictions, which invented a muon-angle anomaly.

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1 How to read this document

Each entry states what was wrong, how it was found, what changed numerically, and what, if anything, had to be withdrawn. "Before" values are quoted from the draft in which they appeared and should not be used; they are given so that an older table can be recognised.

Entries are ordered by the stage of the analysis they affect, not chronologically: exposure and flux first, then the Monte Carlo weighting, then the unfolding, then the systematic covariance.

2 Exposure, flux, and target normalisation

2.1 Obsolete integrated flux (factor $\sim$3.48)

An outdated flux file was used to convert the extracted event rate into a cross section. The integrated flux it supplied was larger than the correct value by a factor of approximately 3.48, so every cross section produced with it was smaller than the truth by the same factor.

Effect. All cross sections, all configurations. Any absolute normalisation from a draft predating this fix is wrong by $\sim$3.48 and must not be quoted. Shapes were unaffected, because the error was a single multiplicative constant.

2.2 Configuration-specific flux was not being used

The RHC and combined extractions were normalised with the FHC integrated flux. The three exposures see different spectra, and the correct integrated fluxes are

$$\Phi_{\mathrm{FHC}} = 6.81159 \times 10^{-10}, \quad \Phi_{\mathrm{RHC}} = 6.44646 \times 10^{-10}, \quad \Phi_{\mathrm{comb}} = 6.60865 \times 10^{-10}$$

in $\mathrm{cm}^{-2}\,\mathrm{POT}^{-1}$.

Effect. RHC cross sections rose by 5.7% and combined by 3.1%. FHC was already correct and did not move. The flux is now set by an explicit `Flux` line in each cross-section configuration file rather than inherited, and `norm_manifest.py` checks that all configurations of a given exposure agree on it.

2.3 Per-mode rather than per-run exposure weighting

Data and Monte Carlo exposures were pooled per horn mode and a single scale factor applied. Because the data-to-Monte-Carlo exposure ratio differs substantially between runs, this weighted the result by how much simulation happened to exist for each run rather than by the data actually collected. Each run is now scaled by its own $S_r = \mathrm{POT}_r^{\mathrm{data}}/\mathrm{POT}_r^{\mathrm{MC}}$.

Effect. Changed the relative weight of runs within every configuration. This is also the change that makes the framework ready for real data, where the per-run exposures are not free parameters.

2.4 Duplicated run-total POT in split files

Two conventions coexist in the processed samples. In `processed/` a run split across several files repeats the *run-total* exposure in every file; in `processed/w/` each file carries its own *true* exposure. Dividing by the per-file value broke the first convention and summing over files broke the second: the first made the inclusive tune come out a factor of two low, the second drove an RHC proton-tagged cross section negative.

The resolution is to normalise by the sum of *distinct* per-run exposures, which is correct under both conventions.

Effect. At the time of the fix the RHC $W_{\pi p}$ cross section moved from -0.194 to $+0.578$ (in units of $10^{-38}\,\mathrm{cm}^2/\mathrm{Ar}$), and the inclusive RHC p_μ result was unchanged at 0.860. Results

retracted on the strength of the negative value were reinstated. Both numbers have since moved again under the central-value weighting fix below; the current values are in the analysis note.

3 Monte Carlo weighting

3.1 Central-value weight applied twice to the fake data

This is the largest single correction and it affected every result.

The fake data are generated externally by `throw_perrun_*.C`, which folds the tuned central value, the PPFX central value and the normalisation into the event multiplicity and then writes the per-event weights back as unity: the throw is a Poisson realisation of an already-weighted expectation. The framework's `build_universes` nevertheless substituted the *weighted* central-value histogram for the raw counts, on both the reconstructed and the truth side, so the central value was applied a second time.

How it was found. By asking directly whether the unfolded result was being compared against the input that generated it. Four independent checks confirmed the diagnosis: 1621.0 is a direct count of selected events across the four FHC files; 1721.9 equals both `FakeDataMC_0_reco` and `onBNB_reco` − `extBNB_reco`; run 1 gives 558.0 entries unweighted against 599.9 weighted for the same 558 events; and the throw's own expectation of 1567.1 agrees with the thrown count at 1.3σ.

Effect. The mean inclusive closure ratio moved from 1.106 to 1.001 (1.016 once the truth-side correction below is included, 1.018 on the re-thrown fake data), and all 51 extracted cross sections fell by between 3.8% and 12.9%. Every absolute cross section from a draft predating this fix is high by roughly a tenth.

Withdrawn. Two earlier explanations of the closure excess, one attributing it to the unfolding covariance and Wiener filter and one to A_C smearing, were both wrong and are retracted. They are retained in the technical supplement as recorded dead ends rather than deleted.

3.2 Generator prediction histograms truncated at the overflow

Generator predictions were read without their overflow contents, so the highest-momentum region was silently missing from the comparison curves.

Effect. Model comparison only; the measured cross sections were unaffected. Any generator χ^2 from a draft predating the regeneration is invalid.

4 Unfolding

4.1 Second-derivative regularisation on a non-uniform grid

The Wiener-SVD second-derivative regulariser used the uniform-grid $(1, -2, 1)$ stencil on bins whose widths differ by factors of ten to fifteen. On such a grid that operator is not an approximation to the second derivative of anything, and its near-null direction is not the physically flat one. The Wiener filter can then retain that direction alone, which drives the additional-smearing matrix A_C towards rank one and destroys the shape information the measurement is supposed to carry.

The corrected operator differences the *density* on a non-uniform grid, reducing to the old stencil when the widths are equal, with one-sided first-derivative rows at the ends so that the operator stays non-singular without asserting that the distribution vanishes at the edges, which it does not, since $\cos\theta_\mu$ peaks at the upper edge.

Effect on the conditioning measure s_2/s_1 of A_C:

Observable	before	after
FHC θ_μ	0.015	0.51
combined θ_μ	0.045	0.38
proton-tagged $W_{\pi p}$	0.022	0.60
proton-tagged p_n	0.21	0.88

Withdrawn. The earlier conclusion that θ_μ "repairs" what $\cos\theta_\mu$ cannot is retracted: under the defective operator every model was being collapsed onto a common shape, so the apparent agreement was an artefact of the regularisation rather than a property of the data. After the fix θ_μ discriminates between models again and prefers the tune that generated the fake data.

This entry is the clearest illustration of why closure alone is not a sufficient validation. A rank-one extraction closes perfectly against its own correspondingly smeared truth while retaining almost no shape information.

4.2 Angular and transverse observables were defined about detector z

The NuMI neutrinos arrive approximately 28.6° away from the detector z axis, but reconstructed angles and transverse kinematic imbalance were being computed about z while the generator predictions were computed about the neutrino direction. These are not the same observables. Truth-level quantities now use the per-event neutrino direction and reconstruction uses the target-to-vertex direction; the two constructions differ only at the milliradian level.

Effect. 64.6% of signal events change $\cos\theta_\mu$ bin and 44.6% change $\cos\theta_\pi$ bin. The mean δp_T moves from 0.742 to 0.264 GeV/c. The muon and pion momenta, the opening angle $\theta_{\mu\pi}$, the hadronic mass $W_{\pi p}$, and the event selection are all unaffected.

Withdrawn. All proton-tagged transverse-kinematic model comparisons made before this correction are withdrawn, because the measured and predicted quantities were different observables.

4.3 Predictions were smeared by A_C twice

Every model prediction plotted or compared against the data was multiplied by the additional-smearing matrix twice: once by the extractor, which applies A_C to each entry of the prediction map in place, and again on the plotting path. Confirmed exactly, as $A_C^2 \cdot \text{raw} = \text{plotted} \times \text{width}$ to 1.0000 in every bin.

How it was found. By testing whether the released A_C lets an external user reproduce a published generator curve. It did not: the discrepancy was 31% on the integral and varied from 1.21 to 1.55 across bins, so it was not a normalisation, and the residual factor turned out to be A_C itself.

Effect (2026-08-31 pass; the current values are in the note's χ^2 tables). Model χ^2 values only. The anomaly the note previously attributed to A_C suppression of sharply peaked generators was this bug:

Observable	Model	before	after
FHC θ_μ	NEUT	40.60/5	8.96/5
FHC θ_μ	NuWro	38.62/5	9.43/5
FHC $\cos\theta_\mu$	NEUT	29.78/5	4.90/5
FHC $\cos\theta_\mu$	NuWro	29.10/5	4.93/5
FHC p_μ	GENIE	5.52/7	0.57/7

Not affected. The unfolded cross sections (FHC p_μ integrates to 0.8275 before and after) and the closure test, because the fake-data truth was the one curve not passed through the

second multiplication. The previous table also predated the central-value weighting fix, so part of the shift in its "truth" column is that correction rather than this one.

Withdrawn. The argument for preferring θ over $\cos\theta$ on the grounds that the $\cos\theta_\mu$ χ^2 was an artefact. The effect that motivated it does not exist.

5 Systematic covariance

5.1 Combined detector-variation exposure double-counted

The combined configuration pooled detector-variation samples per horn mode, and the denominator used a single file's exposure rather than the summed Monte Carlo exposure over all files of that variation type, so the combined detector systematic was inflated.

Effect. The total detector systematic fell from 42.3% to 24.6%, and the combined result moved from 0.910 to 1.027 times its expected value. FHC and RHC were bit-identical before and after, which is the check that isolates the defect to the combination step.

5.2 Proton-tagged detector systematics were absent entirely

Of the fifty-one extractions, the thirty-three proton-tagged ones carried *no* detector systematic at all: the `detVar` column was `n/a` for every proton-tagged observable in every configuration, so their quoted uncertainties omitted a term worth 17–43% in the corresponding inclusive results.

The cause was a configuration gap rather than missing data. The proton-tagged detector variation samples had been processed on 2026-08-20/21 and carry the required branches, but nothing referenced them: the `_w` file-properties tables contained no `detVar` lines, and `ccpi1p_systcalc_numi.conf` had no `DV` block at all. Both were needed; adding either alone does nothing.

Effect (values of the time). The thirty-three proton-tagged universes were rebuilt with the nine detector variation samples included and re-unfolded from them. Every extraction in the analysis now carries a detector term. The proton-tagged terms ran 10.5%–64.4% with a median of 23.9%, against 16.9%–42.8% for the inclusive results (release of 2026-09-26: 9.9–74.5%, median 30.9%, and 16.6–34.2%).

Check (of the time). Per-mode detector-variation double counting has occurred in this analysis before, and the combined configuration, which carries two per-mode sets where the single-mode configurations carry one, is the one exposed to it. Double counting inflates the combined result. On that pass, of the eight observables (of eleven) whose combined value fell outside its own FHC–RHC range, six fell *below* both and only two above, the finite-sample averaging-down expected of statistically limited detector-variation samples rather than a recurrence; on the current release only two of eleven fall outside the range, both above (analysis note).

6 Count-level normalisation, fake data and release repairs (2026-09-02 onwards)

6.1 Cut-and-count uncertainty quoted from a pre-rebuild breakdown

The total flux-averaged cross sections of the cut-and-count path took their systematic fraction from the p_μ systematic-breakdown tables, which had not been regenerated after the detector-variation rebuild of 2026-08-31 (their detector terms were $\sim 10\%$ against 17–43% in the release). The breakdown tables in the note and the supplement were regenerated from the release systematics dumps on 2026-09-02 and the cut-and-count uncertainties recomputed from them; the macro also now applies the central-value MC weight, which moved the central values by 0.3–0.5%.

Effect (values of the time; the current totals are in the one-bin entry below). Inclusive totals (10^{-38} cm^2/Ar): FHC $1.06 \pm 0.27 \to 1.03 \pm 0.43$, RHC $0.98 \pm 0.35 \to 0.93 \pm 0.42$, combined

$1.02 \pm 0.30 \rightarrow 0.98 \pm 0.40$. The generator comparison changes from "GENIE $\sim 1\sigma$ low" to "GENIE 0.5–0.6σ low". A proton-tagged cut-and-count total is quoted for the first time (FHC 0.63 ± 0.31, RHC 0.55 ± 0.21, combined 0.59 ± 0.25).

6.2 Fake-data truth still carried the central-value weight

The 2026-08-30 correction (above) stopped the tune weight being applied a second time to the externally thrown fake data on the reconstructed side. The truth side was missed: the throw macro resets only the scalar weight branches, the `weight_TunedCentralValue_UBGenie` vector survives, so the universe maker still writes a tune-weighted "CV" universe for each fake-data file, and the systematics calculator took that weighted histogram as the fake-data *truth* whenever it was present. Measured on the FHC p_μ universe file: weighted/unweighted true-signal integrals 1.023, 1.008, 1.008, 1.018 for Runs 1, 2, 4, 5, about 1.6% overall. Every closure ratio was therefore low by that amount and every closure χ^2 slightly too small. Found in the 2026-09-02 code trace; the truth path now uses the same external-throw flag as the reco path, and the universe cache key was bumped so the stale sums cannot be reused.

Effect. Unfolded cross sections and covariances are unchanged (FHC p_μ integrates to 0.8275 before and after). Closure: FHC p_μ unfolded/truth $1.024 \rightarrow 1.041$, χ^2 $0.10/7 \rightarrow 0.13/7$; the mean inclusive closure ratio $1.001 \rightarrow 1.016$. All fifty-one sidecars, the released curves (`truth_smeared` column), the closure tables and the closure figures were regenerated; the ensemble bias is unaffected because its reference is the fixed central-value truth, not the thrown one.

6.3 Beam-off gate count was an event count

Every live file-properties configuration scaled the pooled Run-1+Run-3b beam-off sample by bnb_trigs[run]/3 821 593. The denominator was inherited from the earlier custom selection, where it was the sum $610\,496 + 3\,211\,097$ of the two samples' *event* counts; their gate counts, from the analyser's trigger table supplied on 2026-09-02, are $4\,582\,248.27 + 32\,649\,128.65 = 37\,231\,376.92$, a factor 9.74 larger. The selected cosmic rate per gate exposed it: 5.8 per 10^6 gates for the pooled sample against 0.37–1.3 for the dedicated per-period files, versus 0.59 with the gate count. Corrected in all six live configurations and in the counting and cut-flow macros; the universes were re-summed (the cache key tracks the file-properties contents) and all fifty-one extractions, covariances and the release were regenerated.

Effect. EXT background at the FHC final cut $128 \rightarrow 13$ events (RHC $133 \rightarrow 14$, combined $261 \rightarrow 27$); purity $55.8 \rightarrow 59.7\%$ (FHC). Fake-data central values are unchanged by construction (the same EXT is added and subtracted); the EXT-statistics and data-statistics covariance terms shrink, which moves the Wiener-SVD results only through the regularisation (see the regenerated tables). A real-data extraction with the old value would have over-subtracted ≈ 115 events, $\approx 12\%$ of the FHC signal. **Withdrawn:** every beam-off yield, purity and EXT-statistics term quoted in drafts before 1.2. The sideband tables were regenerated with the per-run treatment (`regen_sidebands.sh`); the cosmic control region, formerly EXT-dominated by 4.5 : 1, now has 0.7 EXT events per neutrino-MC event.

6.4 Beam-off matched by run period instead of one pooled sample

Beam-off data are cosmic-only, so a sample is matched to a beam-on run by its run period, irrespective of the horn polarity it was recorded under (analyser, 2026-09-02). The pooled Run-1+Run-3b sample was replaced by the dedicated per-period samples (Run 1; Run 3b; Run 4a, 4b, 4c, 4d pooled; Run 5; the Run-1 sample standing in for Run 2), each scaled by its own gate count from the analyser's table, and processed with the swtrig fail-open build so that the CRT-era samples select cosmics at all. The selected rate per 10^6 gates is 1.75 (Run 1), 0.43 (Run 3b), 1.15 (Run 4), 0.52 (Run 5), so period matching moves the prediction by a factor of about two

relative to the pooled sample. All fifty-one universe files were rebuilt (SLURM array 3205007, 51/51), the extractions re-unfolded, covariances re-harvested and the release regenerated.

Effect. EXT background at the final cut (FHC / RHC / combined): pooled 13/14/27 → per-run 30/23/53 events, i.e. 1.9% of the selected sample; purity 59.1% (FHC). EXT-statistics term on FHC p_μ 0.4 → 0.9%. Unfolded central values move by $\lesssim 1\%$ through the regularisation.

6.5 PPFX central-value weight absent from the cut-flow and cut-and-count counts

The framework's central-value event weight for NuMI is tune × PPFX CV × normalisation (`UniverseMaker.cxx`), and the per-run fake data are thrown with that product. The selection's cut-flow histograms and the cut-and-count macro carried the tune only. The PPFX central-value weight averages 0.944 on selected signal, so every absolute yield in the cut-flow table and every cut-and-count total sat 3–6% above the prediction the unfolding uses; efficiencies and purities, being ratios, were unaffected at the 0.5% level. Found in the 2026-09-02 code trace.

Effect. Cut-and-count totals: FHC 1.06 → 1.03, RHC 0.98 → 0.93, combined 1.02 → 0.98 (10^{-38} cm^2/Ar), and the RHC/FHC ratio 0.928 → 0.901. The selection now fills the cut-flow histograms with the full weight and the fourteen overlay files and the dirt overlay were reprocessed on 2026-09-03 (`reprocess_cutflow_cf.sh`, into a separate directory so the universes' POT convention is untouched); the cut-flow table is regenerated from them. No unfolded result is affected.

6.6 Proton-tagged TKI figure drawn from the withdrawn fine-binned sidecars

The proton-tagged W/TKI montage macros read closure sidecars keyed `dpt`, `dalphat`, `dphit`, `pn`: the withdrawn fine-binned extraction of 2026-08-18, which predates the beam-axis correction, rather than the released two-bin `*2bin` files. The figure therefore compared transverse quantities about detector z (its "data" and tune) with beam-frame generator curves, and showed shape disagreements of a factor of several that do not exist in the release. The staleness check maps figures by configuration only and could not catch it. Macros repointed, the stale sidecars quarantined, the figure regenerated (2026-09-03).

Effect. Figure only; no released number. In the two-bin scheme the generators differ by normalisation (×1.6 GENIE to NEUT) with shapes consistent with the data.

6.7 Result tables rewritten into the wrong place by the table patcher

The script that repatched the result tables after each regeneration located a table by the `\begin{tabular}` nearest its label, falling back to the *preceding* tabular whenever the caption between label and tabular exceeded 400 characters. Four tables were affected in the committed Draft 1.2: the FHC results table received the RHC rows, the RHC table the combined rows, the combined table the rows of the integrated-cross-section summary, and in the supplement the cut-and-count comparison was overwritten by an FHC systematics breakdown. Caught by the second external review (2026-09-03). All release-derived tables are now regenerated as whole tabulars by `rebuild_tables.py` with a locator that requires the tabular to follow the label inside the same environment, and `report/check_tables.py` asserts, on every build, that each table sits under its own label, has the column count its header declares, and agrees with `current_results.tsv`.

Effect. Presentation only; the release files were correct throughout. The tables of the 2026-09-03 build are the first to pass the automated check.

6.8 Two central-value weight rules were in use; the fake data skipped events the prediction counted

The framework's `safe_weight()` sets a non-finite, negative or > 30 central-value weight to unity. The count-level macros (cut-and-count, cut-flow, sidebands, transfer factors) instead zeroed invalid weights and did not clamp, and the per-run fake-data throws *skipped* events with a non-finite weight. About 1.15% of overlay events carry a non-finite tune weight, concentrated in the non-signal ν_μCC classes, so the pseudo-data sat below the framework's own prediction by 0.9% in the signal region (13–14 events) and by 2.8–3.4% in the CC0π control region (341 FHC, 422 RHC POT-scaled events); the sideband yields of the note differed from the systematics tables by the same amounts. Found through the third external review (2026-09-03).

Correction. One rule, the safe-weight definition in the Monte Carlo section of the note, is now applied everywhere: the selection's cut-flow histograms (reprocessed), every count-level macro, and the fake-data throws, which were re-thrown with the same seeds and the events kept at unit weight; all fifty-one universe files were rebuilt from the re-thrown fake data. **Effect.** Pseudo-data yields rise by 0.9% (signal region) to 3.4% (CC0π). Because the re-throw is a new Poisson realisation, the individual Wiener-SVD integrals moved by up to 20% (RHC p_μ $0.573 \to 0.690$) within their statistical uncertainties, the inclusive closure mean moved from 1.016 to 1.018 (all $p > 0.9$), and the prediction-total uncertainty that the cut-and-count totals inherit from the p_μ extraction moved from 44.4% to 37.7% (RHC) and from 37.0% to 53.5% (proton-tagged RHC); the cut-and-count central values, which use the central-value prediction rather than the throw, are unchanged. The D'Agostini cross-check was re-run on the new release: its integral is 20–29% above Wiener-SVD and flat across observables to $\leq 1.0\%$ (previously 27–33% and $\leq 2.1\%$). The real-data path was never affected.

6.9 The total cross section borrowed its systematic from the p_μ extraction

The headline total was a cut-and-count evaluation whose systematic fraction was taken from the prediction total of the released p_μ extraction of the same configuration. That proxy carried the regularisation amplification of the flux and detector terms, and it depended on the fake-data throw through the unfolding operator: on the re-throw of 2026-09-03 alone it moved from 44.4% to 37.7% (RHC) and from 37.0% to 53.5% (proton-tagged RHC) with no change to the central value.

Correction. The total is now extracted with the framework as a one-bin distribution: one true bin containing every signal event, defined on $\cos\theta_\mu$ over its full range so that no overflow exists, one reconstructed bin, Wiener filter off. The response is the efficiency alone, nothing is regularised, $A_C \equiv 1$ exactly, and the covariance is propagated universe by universe like every other extraction. Six new universe files (inclusive and proton-tagged $\times$ FHC/RHC/combined; SLURM 3228639) and `data_release/total_xsec.tsv`. The cut-and-count macro is kept as the closed-form check and reproduces the one-bin extraction of the central-value prediction to four digits. **Effect.** Inclusive totals 1.03 ± 0.40, 0.93 ± 0.35, 0.97 ± 0.37 (previously ± 0.42, ± 0.35, ± 0.34 with the borrowed systematic); proton-tagged 0.62 ± 0.26, 0.53 ± 0.25, 0.57 ± 0.24 (previously 0.63 ± 0.32, 0.55 ± 0.30, 0.59 ± 0.28). The central values now fluctuate with the throw (closure 0.98–1.00 against the realised truth) instead of being the injected tune by construction. The generator comparison moves from "0.5–0.6σ above GENIE, within 0.7σ of the rest" to "0.5–0.8σ above GENIE, within 0.8σ of the rest".

7 Development-era corrections

These predate the framework-based analysis and are recorded for completeness. They were previously listed in a "Known Issues" table in the analysis note; that table had become thirteen

resolved rows against one open one, so it was retired and its contents moved here, where a resolved correction belongs.

Item	Resolution
Unfolding covariance was statistics-only	Replaced by the framework `PredTotal` plus the data Poisson term, removing a ∼26% bias in the quoted uncertainty.
$\theta_{\mu\pi}$ binning mismatch	A custom 10-bin scheme was aligned to the framework's 9-bin one; the two pipelines had been binning the same observable differently.
Negative tail bins	An artefact of the EXT treatment; all bins positive once the subtraction was corrected.
EXT subtraction on fake data	The custom pipeline cancelled it via `fake_data_ext_cancel`, a flag that had to be remembered at unblinding. Superseded: the framework selects the beam-off treatment from the presence of Monte Carlo truth in the beam-on file (added to pseudo-data, subtracted from real data) and refuses a real file unless `XSEC_UNBLIND=1`, so no manual switch remains.
Multisim propagation in the custom pipeline	Never completed, and now moot: the released results come from the framework, which propagates the multisims (`cov_xsec_multi` is in the release). The row survived in the note as the single "open" item long after it had ceased to describe anything.
CRT veto branches	`crthitpe` and `crtveto` are all-zero in the NuMI ntuples; replaced by `bdt_cosmic`, disabled at its default.

8 Review passes

8.1 Corrections from the independent review of 2026-09-04

Two reviewers with no prior contact with the analysis read Draft 1.4 in full, one checking every number against the release files and one as a referee. What they found, and what changed:

Release index. `data_release/index_extractions.tsv` and `README.md` were the 2026-08-31 files and disagreed with every other release artefact; regenerated from the current results (`report/tools/index_extractions.py`), with the differential $W_{\pi p}$ shape flagged withdrawn.

Proton-tagged cut-flow. The supplement's proton-tagged cut-flow table predated the beam-off gate and PPFX-weight corrections, and the note's proton-tagged prose quoted it (47–48% purity, 33 → 48%). The fourteen overlays and the dirt overlay were reprocessed with the proton-tagged selection under the unified weight rule (`processed/cf_1p`); the macro's final stage, which had counted events unweighted (2.6% high), now carries the same weight as the other stages, and its final row reproduces the cut-and-count check exactly (376.4 signal, 723.0 predicted, FHC). Purity is 52% (FHC), 50% (RHC), 51% (combined); the tag rejects 67% of the neutrino background for a 32% signal loss and reduces the beam-off by 3.3–3.5 at the final cut, not six.

Flux term. The note attributed the 24–33% unfolded flux term to amplification by the unfolding operator. The one-bin total, which has no regularisation, carries 28–30%: a flux universe scales signal and background together, so the term is $\delta\Phi/\Phi \times (1 + B/S)$, $17.7\% \times 1.71 = 30.2\%$ against the one-bin 30.25% (FHC). The section is rewritten; the reconstructed-level table of the earlier pass is dropped.

Control-region procedure. The target class of the π^0 region as tabulated (ν_μCC with a π^0) is 46% of the region, below the 50% purity the procedure requires for a fitted correction; stated, with the CC+NC π^0 definition to be ratified. The pass/fail criterion is now a global normalisation χ^2 over the eight (region, mode) totals; the sixteen independent 2σ and $p > 0.05$ tests of the earlier draft (which would have flagged at least one region by chance more often than not) are diagnostics, reported with pulls against the statistical+detector covariance and a normalisation-marginalised shape test. The response to a localised reconstruction failure, the

joint FHC/RHC treatment and the leverage of the held-fixed signal contamination are specified. A control-region skim that never writes signal-region events is added to the prerequisites; it did not exist at that review and was implemented on 2026-09-06 (below).

Adopted 2026-09-06, pending the conveners' ratification. The analysers adopted the π^0 region's target class as CC+NC π^0 (purity 50.1% FHC, 51.0% RHC of the neutrino MC; $T = 0.019$–0.020; transfer uncertainty 17–19%, floor-limited); the class was added to the transfer, transfer-uncertainty and detector-variation macros and to the note's tables, and the purity denominator of the procedure is now defined as the neutrino-MC prediction.

Protocol statistics (2026-09-06). The three statistics of the frozen procedure (global 8×8 normalisation χ^2, per-region pulls against the full and the statistical+detector covariance, normalisation-marginalised shape tests) were implemented (`macros/sb_protocol.C`) and exercised on the full-stack pseudo-data: χ^2 0.66/8, pulls within ± 0.5, shape $p > 0.47$; power 12% (CC0π, π^0) and 50–62% (multi-π, cosmic) for a single-region shift, 58% for a common shift. The response tree was rewritten as an explicit flow (trigger, order of regions, numeric criterion for a localised reconstruction failure, size of the added uncertainty, leverage of the held-fixed components in a fitted correction, post-treatment criterion).

Detector term statistics. The Monte-Carlo-statistical floor of the detector-variation term was measured on split samples (`macros/sb_detvar_split.C`): the CC0π transfer term (27% FHC, 31% RHC) is genuine (floor 9%, 6%); the π^0 transfer term (15%, 11%) is at or below its floor (13%, 17%) and is an upper limit; the region-yield terms have floors below 2%. The procedure now uses the larger of the term and its floor.

Ensemble. The 32-throw ensemble was thrown by a macro that still skipped non-finite-weight events, which accounts for $\approx 1.4\%$ of its 2.5% negative offset. The macro now applies the unified rule and the ensemble was re-run for p_μ, $\cos\theta_\mu$ and the one-bin total (SLURM 3277674, 3277691, 3277692; statistical covariances re-harvested with `slurm/ens_statcov.sh`, pulls with `report/tools/ensemble_stat_pulls.py`). Statistics-only pull widths 1.08, 0.97, 1.02 (previously 1.04, 0.96); ensemble means below the reference by 0.5%, 2.7% and 1.0% at 0.4, 1.9 and 1.3σ (previously 2.5% at 2.1σ for p_μ). The bias statement in the note is reworded accordingly: a small common negative offset, not significant, largest in the most regularised observable, and present at the 1% level in the unregularised total.

Also. The A_C claims made about proton-tagged closure ratios (which are ratios to the A_C-smeared truth) are removed; the RHC-below-FHC comparison of A_C-suppressed integrals is replaced by the one-bin ratio; the reproducibility claim is scoped to the 2026-09-02 pass; the D'Agostini cross-check is stated to be prior-dominated on central-value input; the absence of an alternative-generator fake-data test on the released chain, of an MCS momentum systematic and of gate-ratio and Run-2 stand-in beam-off terms is stated in the Limitations; the signal definition (one charged pion of any momentum, above threshold), the unfolder provenance, the regularisation matrix and the proton-tag values (LLR < 0.05, $p_p > 0.3$ GeV/c) are stated precisely; the flux mean energies (0.48 GeV) and the fraction above 0.3 GeV (40%) are given, which is the origin of the factor ~ 3 against the BNB total. Some twenty further inline numbers that had survived from older passes were corrected.

8.2 Inclusive $\cos\theta_\pi$ re-binned to five forward-split bins

The binning re-check of 2026-09-05 (technical supplement) found that every released binning except $\cos\theta_\pi$ is at its resolution limit, and that $\cos\theta_\pi$ admits a fifth bin in the forward hemisphere. Adopted on 2026-09-06: edges $-1, -0.1, 0.35, 0.6, 0.8, 1$ replace $-1, -0.1, 0.35, 0.75, 1$ for the three inclusive extractions (the proton-tagged $\cos\theta_\pi$ keeps four bins). The four generator predictions were regenerated at the new edges from the event-level files (`generator_predictions/costhpi5`); the unchanged observables reproduce the previous prediction files to machine precision. The three universe files were rebuilt (SLURM 3277974), all extractions re-unfolded and the release regenerated; the D'Agostini cross-check was re-run.

Effect. FHC $\cos\theta_\pi$: σ_{int} 0.726 → 0.674, closure χ^2 0.80/4 → 0.68/5, A_C row sums 0.71–0.82 → 0.68–0.77, $s_{\min}/s_1$ 0.27 → 0.46; RHC 0.663 → 0.615, 0.43/4 → 0.49/5; combined 0.753 → 0.658, 1.05/4 → 0.73/5. The inclusive closure mean moves from 1.018 to 1.010, the range of inclusive integrals to 0.62–0.96, and the D'Agostini-to-Wiener-SVD ratio to 1.23–1.31 (flat to $\leq 1.2\%$). No other extraction changes.

8.3 Documentation-only corrections from the 2026-09-02 review

None of the following moved a released number; they are recorded because a reader of Draft 1.1 would otherwise take the printed formulas as the implementation. (i) The differential cross-section equation divided by the efficiency in addition to unfolding with the smearceptance matrix $S = \varepsilon M$; the implementation applies the correction once, through S, and the equation now says so. (ii) The response-matrix figure was described as the column-normalised migration matrix; it is S, with columns summing to the efficiency. (iii) The muon-candidate cut was described as "longest qualifying track"; the code orders the qualifying set by the muon-BDT score. (iv) Coherent pion production was described as a background; the signal definition is topological and includes it. (v) The single-exposure normalisation example (7.63×10^{20} POT) and a pre-rebuild systematic-breakdown table were replaced by the final three-configuration values. (vi) Flux composition was labelled as integrated above 0.1 GeV; the values are for 60 MeV. (vii) TKI angle units in the binning appendix read rad; they are degrees. (viii) A θ_π result was listed as final; none is extracted. (ix) The beamline-geometry flux term, quoted at 1.6–3.3%, is not in the released systematics configuration and is now stated as absent.

8.4 External review of Draft 1.5: the two blocking safeguards implemented

An external reviewer read Draft 1.5 and the supplement together as a package for opening the inclusive control regions and judged the science mature but the package not yet approvable, on two explicitly required safeguards that the note itself declared missing, plus governance and consistency items. Acted on 2026-09-06: (i) **The signal-region-stripped beam-on skim now exists.** `ProcessNTuples` with `XSEC_CR_SKIM=1` writes an event only if it fails the signal-region selection and belongs to at least one control region, suppresses the selection summary, and stamps the file with a `XSEC_CR_SKIM` marker that `UniverseMaker` refuses regardless of `XSEC_UNBLIND`; the sideband macros refuse any beam-on file that has neither the marker nor Monte Carlo truth (`macros/sb_guard.h`). Verified by `macros/cr_skim_check.C` on three inputs, the Run-1 FHC overlay and the Run-1 and Run-4c beam-off samples, each processed both ways (recorded in the repository at the tag). No beam-on file was touched. (ii) **The π^0-definition comparison** with and without the final multiplicity cut, promised before ratification, was produced from a Monte-Carlo-only reprocess into a separate tree (`/data/uboone/processed/sb_pi0`, SLURM 3277998; new branches `sb_pi0_final`, `sb_nprimtrk`, `sb_nnonproton`); results in the supplement, §20.4: the cut removes 11–14% of the region uniformly across its components, and the frozen definition is kept. (iii) **Governance.** The run-period rule no longer excludes a period automatically on a 3σ excursion: it triggers an investigation, exclusion requires an independent detector or processing defect and a new frozen release. A fitted π^0 correction is not pre-authorised by the 50.1/51.0% purity: it may be proposed only after the pre-fit result is documented and reviewed. What a fitted correction updates in the covariance is now stated (target-class normalisation only, re-evaluated per universe; flux coherence retained). The two fitted regions are disjoint by construction, so no joint fit counts an event twice. (iv) **Consistency.** The supplement's two-bin p_π passage still carried a "1.13 in all three" unfolded-to-truth ratio from a superseded pass beside the current 0.997/0.981/0.989; withdrawn, and the A_C diagonals quoted there brought to the release (0.53/0.38/0.53 first-bin diagonals). The note's summary pointed the identity closure to the supplement's cross-check section instead of its identity appendix; fixed. "Full covariance" is now "complete configured covariance" with the identified omissions named beside

it; the supplement no longer claims the two documents are consistent "by construction". A boxed scope statement follows the golden-pion conclusion so it is not read as contradicting the adopted two-bin diagonals; the supplement's normalisation appendix is retitled as a superseded record; the cosmic region is introduced as the large-opening-angle, cosmic-enriched region; "beam-off", "EXT" and "cosmic data" are declared synonyms at first use; "signal contamination" is defined as the inclusive signal unless stated otherwise.

8.5 Second external review (2026-09-07): a stale figure and three captions

The external reviewer re-read the 2026-09-06 note and supplement. The 1.13 contradiction is confirmed resolved. Found and corrected on 2026-09-07: (i) **The proton-tagged two-bin p_π figure of the supplement was stale**: it printed integrals of 0.507/0.447/0.490 against the table's 0.498/0.429/0.462. The producing macro (`macros/ppi2bin_figs.C`) read the 2026-08-30 sidecars `closure_hists_xsec1p_*_ppi.root` while the tables read the current `closure_hists_xsec_ccpi1p_*_ppi2bin.root`; the macro now reads the current sidecars for both families and all nine figures are regenerated (inclusive values unchanged). The staleness guard could not catch this, since it compares modification times and the figure postdated the sidecar; it now also reads the integral printed on each two-bin figure and compares it with the sidecar's. (ii) The proton-tagged closure-table caption said the integral ratios "are the A_C smearing effect"; both numerator and denominator are in A_C-smeared space, so the ratios measure the throw's realisation and any residual bias. Corrected. (iii) The bin-integrated two-bin p_π table called its uncertainty the data-statistical term; it is the total from the released covariance ($1.36 \times 0.030 = 0.041$). Corrected. (iv) The inclusive two-bin figure caption named NEUT "closest"; the closest generator changes by bin. Reworded. (v) The row-sum interpretation is now stated for a flat truth, with the model dependence acknowledged; "retains 5% of a bin's normalisation" became a row sum of 0.05; the supplement's "property of the response, not of the model" is qualified to the tested generators. (vi) The open upper p_π bin: the released density and the conventional width form a matched representation of the bin integral, stated.

8.6 Stale-detail sweep (2026-09-07)

Three independent reviewers read the four documents and the data-release index for statements overtaken by the last week's changes. Nothing numerical moved. Corrected: the skim verification is now described as three inputs everywhere (Run-1 FHC overlay, Run-1 and Run-4c beam-off); the remaining "full covariance" phrases became "complete configured covariance" (note, talk, release README and index header); the beam-on normalisation paragraph of the note now states the distinct-exposure rule; the CRT paragraph no longer says the branches are zero-filled (they are pass-through, populated for Runs 3–5 only); the supplement's proton-tagged A_C mean row sum for W_{had} is 0.61 (release value), the pre-fix $\cos\theta_\mu$ generator-suppression passage and the 2026-08-28 combined-detector and spread passages are labelled with their pass, the binning re-check tables say "released" means the 2026-09-04 release, and the W_{had} edge scan is recorded as open (non-gating; note, Limitations); the talk's backup control-region frames say "adopted target; ratification requested"; the change log is regrouped (count-level repairs and review passes as sections), its superseded table gains the cut-and-count totals, proton-tagged purity, ensemble offsets and FHC beam-off rows, and the note's count of result-affecting entries is twenty (bundles counted once; the three presentation- or documentation-only entries excluded).

8.7 Referee re-review (2026-09-07)

The external referee re-read the package as a whole. Corrected: the closure-χ^2 discussion said the uncertainty sizing was tested by neither a restricted covariance nor an ensemble, contradicting the ensemble subsection that follows (reworded); the provenance sentence intended for the binning

re-check tables had landed twice in the total-cross-section caption (moved); the status matrix now says "complete configured covariance" and marks the proton-tagged differential results as released secondary with binning validation incomplete (W_{had} edge scan not run); the π^0 region's paragraph states the chronology (cut dropped for a plausible sculpting argument; the comparison found none at the composition level; the no-cut definition is kept as the larger sample); the $1 + B/S$ flux mechanism is set in a box; the bin-integrated p_π representation is declared primary. Added, on the referee's suggestion: a null calibration of the global control-region test (2×10^5 pseudo-experiments, prediction from the pre-fit covariance, Poisson data): the nominal $p < 0.05$ threshold fails 5.1% of null experiments and $p < 0.01$ 1.1%, so the χ^2_8 asymptotics hold for this covariance (finite-universe noise not re-thrown).

9 Corrections of 2026-09-16/17 (control regions and fit statistics)

Found while building the joint FHC+RHC background fit; all three are documented in full in the control-region note (its §5 and §6), which carries the corrected tables. The released cross sections are *not* affected by any of them.

9.1 The sideband Monte Carlo predated the beam-frame correction (numbers moved)

The sideband-instrumented Monte Carlo `processed/sb/` was produced 2026-08-11, before the beam-frame correction of the muon angle (2026-08-19), so it stores $\cos\theta_\mu$ about the detector z axis while the beam-on control-region skim stores the beam-frame angle. Every $\cos\theta_\mu$ comparison of the 2026-09-09 control-region note compared two different conventions. Data over prediction in the CC0π region (FHC Runs 4+5) swung from 0.22 to 2.46 across the angle bins; with the current Monte Carlo (`processed/sb_pi0/`, 2026-09-06) it is flat at 1.10–1.34. The shape tests move accordingly:

Normalisation-marginalised shape test	2026-09-09	corrected
CC0π $\cos\theta_\mu$, FHC	350.51/4	3.26/4
CC0π $\cos\theta_\mu$, RHC	583.58/4	5.48/4
π^0 $\cos\theta_\mu$, FHC	210.36/4	0.59/4
π^0 $\cos\theta_\mu$, RHC	240.85/4	0.49/4

The region flags do not depend on the angle, so the yields, the per-run table and the global normalisation test are unchanged. The p_μ shape failures (π^0 in both modes, RHC CC0π) survive the correction. The muon-angle shape failure reported in that note is withdrawn. `sb_plots_data.C`, `sb_perrun_data.C` and `sb_protocol_data.C` now read `sb_pi0/`; the control-region data figures are regenerated (the originals are kept as `sideband_data_*_sb20260811.pdf`).

9.2 Sideband detector variations were not POT-normalised (numbers moved)

The sideband macros formed the detector-variation shifts as ratios of unscaled event counts. The knob samples do not have the POT of the detector-variation central-value sample (FHC: WMX 0.913, LYdown 0.926, Recomb2 1.039), so that ratio carried the exposure difference as a spurious shift of about -8% per knob. The detector term of the control-region covariance:

Detector term (quadrature over 8 knobs)	CC0π	π^0	cosmic	signal region
unscaled ratios, FHC / RHC	20.2 / 8.4%	20.1 / 7.2%	21.7 / 30.9%	21.0 / 15.4%
POT-normalised, event-matched, FHC / RHC	1.9 / 2.1%	0.8 / 1.8%	6.5 / 7.0%	8.3 / 8.5%

With the corrected covariance the frozen global normalisation test moves from $\chi^2 = 8.77/8$ ($p = 0.36$, passes) to 69.5/8 ($p < 10^{-3}$, fails); excluding FHC Run 1, whose exposure is under investigation, it passes again at 5.0/8 ($p = 0.76$). Transfer-factor ratios (numerator and denominator from the same sample) are unaffected, as is the extraction, which normalises by `summed_pot`. The statistical part of the detector variations is now removed with an event-matched Poisson bootstrap: the knob samples are re-simulations of the same generated events (96.8–98.1% matched), so the noise is 3–10× smaller than an independent-sample estimate; in fine bins the median quadrature shift of 8.3% is 7.3% noise. `sb_protocol_data.C` takes both corrections as switches, on by default; with both off it reproduces the published tables exactly.

9.3 Fit statistics: CNP data variance leaves a pull bias (presentation of the new fit only)

In the background fit, 10^5-toy ensembles show the combined Neyman–Pearson data variance, which depends on the observed data, leaves a pull-mean bias of +0.019 to +0.034 in the two-dimensional fits (six to ten ensemble standard deviations); it survives Gaussian toys, so it is not Poisson counting. With the Pearson variance at the nominal prediction the pull mean converges to zero within 0.005. The fit default is the Pearson form. No released number is involved.

9.4 New documentation

`report/newobs_note.tex` records the 0.50 migration-diagonal study (no rebinning adopted: all twenty-six candidates are rejected by the A_C conditioning), the new proton-tagged observables θ_p and $\theta_{\pi p}$ with their reprocess and its bit-for-bit gate, and the two-dimensional programme. The control-region note gains its §5 (corrections) and §6 (joint FHC+RHC background fit).

10 Corrections and additions of 2026-09-18

10.1 θ_p adopts RHC-derived edges

The maximin binning scan had only ever been run on the FHC chain, because `macros/tki_binning_scan.C` chained the FHC files unconditionally; it now takes a beam argument, FHC by default so every published screen reproduces. Run on RHC it returns different edges, and at the *same* four bins those edges condition three to four times better in both usable configurations (FHC $s_{\min}/s_1$ 0.082 → 0.383, combined 0.125 → 0.408). θ_p therefore adopts 0, 0.659, 1.068, 1.539, π rad. Coarsening to three bins does *not* rescue the rank-deficient RHC response (0.002 → 0.050), so θ_p is quoted for FHC and the combined configuration only. The price of the change is stated with it: the data-statistical term rises from 12.3% to 16.6% (FHC) and 9.7% to 14.7% (combined), because better conditioning means less of the truth smeared into the measurement, not a smaller error bar. Configurations, generator predictions and all three extractions were rebuilt; regenerating the predictions left 240 histograms byte-identical and changed only `thetap_fte`. Superseded outputs are kept as `*_thetap_fhcedges`. No released observable is affected.

10.2 The proton candidate, and where its cost actually appears

On selected signal events the reconstructed proton candidate is the leading true proton 81.7% of the time, some true proton 87.4%, and not a proton at all 12.6% (the 59% recorded during the screen predates the proton-tagged reprocess and is withdrawn). A wrong pick randomises the angle rather than smearing it, so it widens the core resolution by only a quarter to a third while supplying 74% of the residuals beyond 0.3 rad in θ_p and 51% in $\theta_{\pi p}$: candidate purity dominates bin-to-bin *migration*, which is the mechanism behind θ_p's marginal diagonal and its rank-deficient RHC response. No released number changes.

10.3 Far sidebands, and a ratio that does not move

Adding one cut per control region drives the CC0π signal fraction from 8.6% to 2.2% while keeping 58% of the events and raising the CC0π content of the background from 71% to 85%. The data-to-prediction ratio is flat under the far cuts (1.12 $\rightarrow$ 1.13; π^0 1.20 $\rightarrow$ 1.23), so the common excess of data over prediction is not signal contamination. *Corrected the same evening:* the first version of this study reported an 8–16% *deficit* because `bkgfit/farsb.py` summed the prediction over the FHC and RHC Run-2 periods, which have simulation but no beam-on file ($\sim$ 20% of the prediction), and because the FHC Run-1 exposure was still wrong; with both fixed the ratios are 1.1–1.2, and the cosmic region falls to unity as its beam-off share rises, locating the excess in the neutrino simulation.

10.4 Corrections to claims in the documents

- The technical supplement stated that the released proton-tagged binnings "were not re-tested". They have been, at both 0.68 and a relaxed 0.50 diagonal: twenty-six candidates pass the relaxed diagonal and all fail on A_C conditioning. Only a two-bin W_{had} is healthy, and it was not adopted.
- The analysis note carried the request to open the control regions with no record that they were opened on 2026-09-09. It now records the outcome, including that the sign-off checklist was not revised at opening.
- `norm_manifest.py` reported a spurious UNFOLD PRESCRIPTION DRIFT and rewrote the tracked manifest on every run: the one-bin total extraction is deliberately unregularised, so the two classes are now audited separately.

11 Corrections of 2026-09-19

11.1 The FHC Run-1 exposure: confirmed and adopted

The FHC Run-1 control-region deficit (fitted exposure factor 0.520 ± 0.020; every region at 0.60–0.72 against unity elsewhere) had a specific cause: the beam-on sample was recorded *without* the open trigger, 2.192×10^{20} POT with 5 748 692 triggers, while every configuration carried the open-trigger pair, 3.283×10^{20} with 9 846 635. Six pairings were tested against the four regions, including a third trigger count found in the repository; only the non-open pair brings all four to unity (0.970, 1.090, 1.380 on 40 events, 0.948), and the beam-off-dominated cosmic region discriminates the gate count on its own. The pairing was then confirmed against the beam bookkeeping.

Adopted in every configuration (17 file-properties lists, the throw macros, the gate counts in the analysis code, the normalisation audit); FHC Run 1 alone re-thrown; FHC exposure $8.857 \rightarrow 7.766 \times 10^{20}$, combined $1.9939 \rightarrow 1.8848 \times 10^{21}$. The FHC and combined extractions are rebuilt on it (universe files, unfolding, covariances, release, figures and tables). On the blind fake data the exposure cancels, because the pseudo-data are thrown at the assumed exposure and numerator and denominator move together, and the rebuilt integrals differ from the previous release only through the new FHC Run-1 Poisson realisation and the changed run weighting (median ratio 0.99; individual bins within their statistical uncertainties). On beam-on data, where the event count is fixed, the same correction raises σ_{FHC} by 14.0% and σ_{comb} by 5.8% relative to the old bookkeeping; RHC is unaffected. The combined flux constant is re-weighted to the corrected POT ($6.60865 \rightarrow 6.59691 \times 10^{-10}$ cm^{-2}/POT). `bkgfit/run1_trigger.py` is the record of the evidence.

11.2 The rebuild on the corrected exposure landed; what moved and what did not

All 34 FHC and combined universe files were rebuilt on the corrected exposure (SLURM 3351878), promoted (the previous files are kept in `univmake_backup_job3351878`), and the full chain re-run: 51 re-unfolds, covariance harvest, release export, figures, counting and every release-derived table. The one-bin totals, which the rebuild manifest had omitted, were rebuilt separately (SLURM 3352400). Because the pseudo-data are thrown at the assumed exposure, the exposure cancels in a fake-data extraction and nothing moved by 14%: the rebuilt integrals are 0.95–1.05 of the previous release for the inclusive observables (median 0.99) and within ±30% in the sparsest proton-tagged bins, all inside their statistical uncertainties, from the new FHC Run-1 Poisson realisation plus the changed run weighting (Run 1 is now 28% of FHC, not 37%). The closure mean is unchanged (1.011 inclusive, 1.012 proton-tagged). The +14% / +5.8% applies to a beam-on extraction at fixed event count, and the documents now say so.

11.3 Three further places the correction had not reached

- **The combined flux constant** was still the 8.857 : 11.082 POT-weighted mean of the FHC and RHC constants (6.60865×10^{-10}); re-weighted to 7.766 : 11.082 it is 6.59691×10^{-10} cm^{-2}/POT (-0.18%), applied to all 50 combined configurations and audited by `norm_manifest.py`; the combined extractions were re-unfolded.
- **Hard-coded per-run scales.** Some twenty standalone macros and tools (cut-flow yields, the counting cross-check, binning screens, every sideband macro, the background-fit templates, `farsb.py`) carry the per-run data/MC POT scale, the full-exposure dirt scale, the beam-on trigger counts or the combined flux as literals. FHC Run 1 was 0.14101 (3.283/23.282) in all of them; it is 0.09415 now, with the dirt scale 0.092402 → 0.081020 where it stands for the full FHC exposure and the pooled-EXT factors of the counting macro recomputed. The framework itself takes all of this from the file lists and was never affected. The cut-flow tables and the counting log are regenerated from the corrected macros; the background-fit templates and the sideband protocol are being re-run on them.
- **Exposure-derived numbers in the documents** (POT tables, integrated flux and conversion factors, the combined $\nu_\mu/\bar{\nu}_\mu$ flux row, the cut-flow headers).

11.4 Background fit and frozen protocol re-run on the corrected exposure

The background-fit templates (`bkgfit_templates.C`) and the frozen control-region protocol (`sb_protocol_data.C`) carried the wrong Run-1 scale; both are re-run. The frozen global test now **passes**, $\chi^2 = 8.1/8$ ($p = 0.42$; it was 69.5/8), and the all-period totals fit needs no parameter (5.7/6; CC0π free 0.949 ± 0.190; the "no FHC Run 1" results are unchanged). A Run-1-specific residual survives: every other period sits 12–25% above the pre-fit prediction, FHC Run 1 at unity, so a free Run-1 factor still fits 0.869 ± 0.022 ($\Delta\chi^2 = 37$; a free factor on any other period gains at most 15). The exposure correction explains 0.52 → 0.87, not 0.52 → 1. Its cause is not identified and it is a new open item for unblinding; the CR note carries the per-period table. The sign of the "residual deficit" statements has changed with the correction: on the corrected exposure the regions sit *above* the prediction, and the far-sideband study is being re-run.

11.5 The MCS momentum-scale bound is withdrawn and redone on the data side

The ±5%/±10% bound quoted on 2026-09-19 scaled the MCS momentum through the *bin configuration*, which applies to the simulation and to the fake data alike. A common scale cancels in the extraction, since response and data move together, so the 0.07σ and 0.19σ measured

only the regulariser's response to a remapped variable, not the systematic. A momentum-scale error is a data-versus-simulation mismatch, so the variation is now applied to the fake data alone (`macros/mcs_scale_fakedata.C`: every MCS-measured muon, 63% of the selected sample, has its momentum scaled by 1 ± 0.05; the simulation, the response and the selection flag are untouched) and $d\sigma/dp_\mu$ is re-extracted with the nominal response (SLURM 3352423). Result: the half-difference is 8–13% of the bin value in the four lowest bins, 2% in the fifth and 8–10% with the opposite sign in the two highest, i.e. up to 0.34σ of the quoted per-bin uncertainty (the withdrawn common-scale test had given 0.07σ); the integral moves -6.8% / $+2.1\%$. As recommended by the review it is released as a fully correlated term for FHC p_μ (`cov_MCSscale.txt`, `cov_total_plusMCS.txt`; per-bin totals 38–58% → 39–58%); propagation inside the framework and to the other configurations and observables is open.

11.6 Proton-tagged alternative-model closure

The reweighted-model fake data (GENIE multisim universe 545; $\Delta \to N\pi$ angular unisim) are thrown for the proton-tagged sample as well (`macros/throw_altmodel_w.C`, same universes and seeds) and $W_{\rm had}$, the four two-bin TKI variables and $\cos\theta_\pi$ are re-extracted with the nominal response (SLURM 3352430), one of the six prerequisites listed for the proton-tagged products. Outcome: nothing beyond 0.9σ under the $\Delta \to N\pi$ variation; under the coherent GENIE variation the low-imbalance δp_T and p_n bins come back 35% below the alternative truth (-1.14σ, -1.16σ) while that truth moved 1%, i.e. the shift enters through the response and background under the varied model. Larger than the inclusive maximum (0.93σ); recorded in the proton-tagged note as a model sensitivity of those bins at the level of their uncertainty.

11.7 Study extractions rebuilt on the corrected exposure (2026-09-20)

The proton-angle products (θ_p, $\theta_{\pi p}$; FHC and combined, live tree), the sixteen FHC and combined two-dimensional pairs and the surviving θ_p binning candidates were built before the Run-1 correction and are rebuilt on it (SLURM 3352599, 24 extractions; the previous live $\theta_p/\theta_{\pi p}$ files are kept in `univmake_backup_study_pre_run1fix`). As for the release, the conditioning is a property of the response and barely moves; the throw-dependent numbers do: $\theta_p \times \delta p_T$ conditions at 0.611 combined (was 0.617), $\cos\theta_\pi \times \cos\theta_\mu$ at 0.564 (0.563), the two-dimensional chain reproduces the one-bin total to a median of 1.5% (0.7%), θ_p conditions at 0.325/0.409 FHC/combined (0.383/0.408), and the three-bin θ_p alternative at 0.12/0.05, combined 0.49. The 2D figures, the proton-angle figure, the inventory and information-content tables and the proton-tagged note's 2D table are regenerated; the conclusions (two pairs usable, combined only; θ_p FHC and combined only) are unchanged. The inclusive FHC alternative-model fake data, thrown on 2026-09-18 at the old exposure, are re-thrown at the corrected one and the eight FHC extractions re-run (SLURM 3352603): the largest bias is now 0.93σ (a sparse $\cos\theta_\mu$ bin under the angular variation; was 0.81σ), the near-threshold p_π region returns 0.40 of the alternative truth (was 0.415) and the sparse top p_μ bin 0.58 (was 1.475: the tail bin is dominated by its own realisation); the populated bins are robust as before.

12 Corrections of 2026-09-24

12.1 Ensemble coverage: ten members rebuilt from truncated fake data (numbers moved)

Completing the six ensembles to 100 members raised the RHC-total statistical pull width from 1.10 (46 members) to 1.28, with the ensemble mean -1.5% (2.9σ) low. The cause was a race condition in the ensemble machinery, not the extraction: the throw file of a member is named by beam mode and throw number only (`ens/fakedata_rhc_run2_t57.root`), and every ensemble

job re-throws it. The combined and RHC-total ensembles started the same throw numbers at the same moment (SLURM 3352020/3352022, 2026-09-20 17:10), so one job read the file while the other was recreating it and silently saw part of the tree. A check of every member (events read by univmake against entries on disk, 2800 inputs) found RHC-total members 56, 57, 59, 60 with Run 2 read at 47–72% and FHC $\cos\theta_\mu$ member 26 at 43%; five further RHC-total inputs differ by one entry, which proved to be an out-of-range truth event in the one-bin binning and not a truncation. The throw macros now write to a temporary file and rename it on completion; the affected members were quarantined (`ens/corrupt_race_2026-09-20/`) and re-run (SLURM 3427769/3427770). Corrected results, all at 100 members: statistical pull widths 0.92–1.07 (RHC total 1.28 → 0.99), 68% coverage 64–74%, ensemble means 0.4–1.1% low at 0.5–2.0σ on the mean (the 2.0σ is the RHC total, −0.81%). The coverage tables of the note and the supplement and `data_release/ensemble_2026-09-20.tsv` are regenerated.

12.2 MCS momentum-scale term extended to RHC and combined p_μ

The data-side variation of 2026-09-19 is now run for the other two configurations (`macros/mcs_scale_fakedata.C` takes the beam mode; RHC fake data ±5% on the 60% of selected muons measured by MCS; the combined configuration scales FHC and RHC together; SLURM 3427764). Combined: half-differences 5–10% in the five lowest bins, 2% and 14% in the two highest, at most 0.35σ; integral −2.5%/+2.4%. RHC: 6–15% ($\leq 0.32\sigma$) in the six lowest bins but +21%/−44% (0.85σ) in the open top bin; integral −0.8%/−6.8%. Released as `cov_MCSscale.txt` and `cov_total_plusMCS.txt` under `incl_RHCFULL_pmu` and `incl_COMB_pmu` (`report/tools/mcs_eval.py CFG`). The RHC top bin is the one place where the term is not small against the quoted uncertainty. Propagation inside the framework and to the other observables remains open.

12.3 MCS momentum-scale term is zero for the other inclusive observables

The same data-side ±5% variation was extracted for $\cos\theta_\mu$, $\cos\theta_\pi$, p_π (two regions) and the one-bin total (SLURM 3427873, stopped after 9 of 36 jobs). Every result is identical to the nominal one (ratio 1.000000 in every bin): the variation changes only the reconstructed muon momentum, which no selection cut and no other inclusive observable uses. The term is therefore exactly zero for all inclusive observables except p_μ. It remains to be evaluated for the proton-tagged transverse-kinematic observables, which use the muon momentum vector.

12.4 Values corrected while rewriting the analysis note (numbers moved)

The analysis note was rewritten in the order of the analysis, without release history, and the following values that the exposure correction of 2026-09-19 had not reached were corrected:

- Combined generator predictions: `combine_comb_fte.C` and `combine_ext.C` weighted the modes with the old FHC exposure (8.857×10^{20}), FHC fraction 0.458 instead of 0.425. The combined generator files were regenerated (their predictions move by −0.6% inclusive and −0.8% proton-tagged; the old files are in `generator_predictions/comb_backup_pot8857/`) and the result set re-unfolded; only the combined generator curves and χ^2 change.
- `data_release/ext_gates.tsv` and the beam-off figure still carried FHC Run 1 at 3.283×10^{20} POT and 9 846 635 gates (FHC beam-off 30.2 events); both are now generated by `report/tools/ext_gates_table.py` from the live file lists (FHC 23.2, RHC 23.2, combined 46.4 events).
- Note: the combined data POT of the exposure table (19.939 → 18.848×10^{20}), the combined flux composition (53.0/44.4 → 52.4/45.1%), the background composition at the final cut (old exposure), the control-region yields and pseudo-data ratios for FHC and combined

(regenerated with `sideband_compare.C`), the two-region p_π table (FHC 0.059/0.764 → 0.037/0.827; now generated by `rebuild_tables.py`), the A_C row-sum table (now generated; FHC $\cos\theta_\mu$ 0.17–0.39 → 0.22–0.42), the flux-term arithmetic ($B/S = 560/833$), the one-bin breakdown ($38.5 \oplus 4.3 \to 38.0 \oplus 4.5\%$), the combined mode weights (0.80:1 → 0.70:1) and the FHC beam-on control-region figure (regenerated on the corrected exposure).

- `tables/infocontent.tex` read hand-copied uncertainties of an older release (p_π 53.9% against 73.0%); it now reads the systematics dumps.
- The glossary defined the migration diagonal as a row-normalised fraction; it is the diagonal of the column-normalised migration matrix, as used throughout.

13 Additions of 2026-09-25

13.1 Statistical coverage of the proton-tagged family

Three 100-member fake-data ensembles now cover the proton-tagged family (SLURM 3427889–91, job script `slurm_ensemble_1p.sbatch`): W_{had} and the two-bin δp_T in FHC, and the two-bin p_n in the combined configuration. The throws are written to a temporary file and renamed, so the race of 2026-09-20 cannot recur, and every member was checked (events read against entries on disk, 1600 inputs, all equal). Statistical pull widths 1.02, 1.08, 0.99; 68% coverage 68.7, 62.0, 64.5%; ensemble-mean integrals +0.02, −0.65, −0.32% from the reference ($\leq 0.8\sigma$ on the mean). The sparse first W_{had} bin is the least regular (mean −0.21, width 1.14), within two standard errors. Prerequisite 3 of the proton-tagged note is met for these three observables (its section "Statistical coverage"). The table is generated by `ensemble_tables_1p.py` and the numbers are released in `ensemble_1p_2026-09-25.tsv`.

14 Binning review of 2026-09-26

14.1 The 0.68 and 0.50 criteria applied uniformly (study; no binning changed)

Every observable of both families was re-binned on the corrected exposure under the nominal 0.68 and a relaxed 0.50 column-normalised migration criterion, applied in FHC and RHC, with an event floor (70 selected FHC signal events per inclusive bin, 50 per proton-tagged bin). At every bin count up to twelve the partition with the largest worst-bin diagonal was found by dynamic programming (`report/tools/maximin_binning.py`), and 141 candidates were extracted in full in FHC, RHC and combined (SLURM 3428213, 3428222, 3428223, 3433027, 3433033). The shape each binning retains is measured by the Wiener filter factors, the eigenvalues of A_C, and their sum, the degrees of freedom for signal; a binning is selected as the fewest bins reaching 90% of the largest admissible sum (`report/tools/binreview_tables.py`, `data_release/binreview_choices_2026-09-26.tsv`). The supplement section (§7) gives the tables, the figure and the observable-by-observable findings. The released p_μ (worst diagonal 0.33) and W_{had} (0.09) fail both criteria; $\cos\theta_\mu$ (0.67) and θ_μ (0.65) fail 0.68 and pass it with optimised edges at the same five bins; no two-bin $\delta\alpha_T$ passes 0.68; the θ_p and $\theta_{\pi p}$ study binnings hold 21 and 42 events in their sparsest bin. Selected under 0.68 and 0.50: p_μ four and six bins, p_π two and three, $\cos\theta_\mu$ seven and eight, θ_μ eight and ten, $\cos\theta_\pi$ seven, $\theta_{\mu\pi}$ five to ten (equivalent), W_{had} two, $W_{\pi p}$ one and two, δp_T and p_n two, $\delta\alpha_T$ one and two, $\delta\phi_T$ two and three, θ_p four, $\theta_{\pi p}$ three and four. The same partition retains up to one degree of freedom more as $\cos\theta_\mu$ than as θ_μ, because the regulariser acts on the density. The review recommended 0.68 for every observable; the decision, taken the same day, is recorded in the next section. A single-throw closure cannot establish the coverage of a new binning, so each adopted one needs its own ensemble.

14.2 Statements corrected by the review (text moved)

Four statements about the released binnings were wrong and are corrected. The note's appendix on alternative binnings said that every binning is the coarsest that meets the criterion and that the conditioning of A_C, not the diagonal, limits the binning: the released p_μ and W_{had} fail both criteria, $\cos\theta_\mu$ and θ_μ fail 0.68, and the threshold of about 0.1 on $s_{\min}/s_1$ used to reject candidates is not met by most released binnings. The note's appendix on distributions not reported said that the W_{had} response is adequate at six bins: all six fail 0.68 and five fail 0.50. The supplement's re-check said that $\cos\theta_\mu$ and θ_μ cannot go finer: with optimised edges both admit more bins, and their released five bins fail 0.68 where optimised five-bin edges pass. The RHC-derived θ_p edges were presented as an improvement: they leave 21 selected events in the last bin. The proton-tagged note's binning prerequisite now states the W_{had} finding, and the note's unblinding requirements include the choice of criterion and binnings.

15 Binnings of the 0.50 criterion adopted (2026-09-26)

15.1 The decision and the binnings

The 0.50 migration criterion was adopted for both families, with the binnings selected by the rule of the review (supplement, §7) and four exceptions: $\cos\theta_\pi$ keeps its five bins, $\theta_{\mu\pi}$ its five, $\delta\alpha_T$ its split at 120°, and the proton-tagged p_π its two regions, since three fail 0.50 in that sample. The binnings are p_μ six bins (0.15, 0.30, 0.425, 0.575, 0.825, 1.25, open) and $\cos\theta_\mu$ eight (-1, 0.4, 0.625, 0.75, 0.825, 0.9, 0.95, 0.975, 1) in both families; the inclusive p_π in three regions (0.175, 0.205, 0.26, open); W_{had} two regions split at 1.18 GeV/c^2; $\delta\phi_T$ three bins (0, 22.5, 75, 180°); θ_p four bins (0, 0.44, 0.785, 1.162, π) and $\theta_{\pi p}$ four (0, 0.911, 1.414, 1.916, π) at their maximin edges. $d\sigma/d\theta_\mu$ is no longer reported: a partition has the same diagonals in θ_μ and $\cos\theta_\mu$, and the cosine retains up to one degree of freedom more. The exemption of p_μ from the criterion ends.

15.2 Proton momentum added to the proton-tagged family

The momentum of the leading proton, p_p, is a new proton-tagged observable. Its candidates, two to five bins between the signal threshold of 0.3 GeV/c and an open top, were extracted in the three configurations (SLURM 3433088); the rule selects three bins split at 0.42 and 0.60 GeV/c (worst diagonal 0.74/0.73, trace 1.49/1.40), and two bins under 0.68. Its generator predictions were built from the event-level files. The release now holds fifty-one extractions: fifteen inclusive and thirty-six proton-tagged.

15.3 Chain

Generator predictions at the new edges were built from the event-level files (`generator_predictions/gen2d`; the histograms of the previous binnings are unchanged and the integrals of every new axis equal those of the old ones). The 27 universe files of the new binnings (SLURM 3433061) and the p_p candidates were installed by `promote_050.sh`, which backed up the replaced configurations and universe files (`configs/backup_pre050_20260926`, `univmake_backup_pre050_20260926`), archived the products of the retired extractions (two-region inclusive p_π, θ_μ, two-bin $\delta\phi_T$) and regenerated the release; the retired figures moved to `figures/retired_pre050_20260926`. The MCS term was re-evaluated for the six p_μ bins (largest 0.56σ, in the open top bin in FHC). The alternative-model closure was repeated for every rebinned observable in FHC (SLURM 3433100, 3433125) and for p_μ, $\cos\theta_\mu$ and p_π in RHC (SLURM 3433342): the largest inclusive bias rose from 0.93σ to 1.18σ (the third FHC $\cos\theta_\mu$ bin under the angular variation, a bin with 114% uncertainty), RHC stays below 0.83σ, and p_p below 0.37σ. Also repeated: the D'Agostini cross-check on the five inclusive observables (integrals flat in the observable to 1% and equal to the one-bin totals), and the five ensembles of rebinned

extractions (FHC p_μ, FHC and RHC $\cos\theta_\mu$, combined p_π, proton-tagged FHC W_{had}; SLURM 3433218, 3433219, 3433290, 3433291, 3433292; 500 members, none failed, no truncated read): statistical pull widths 1.07, 0.97, 1.05, 0.95 and 0.94, integral offsets -0.4 to -1.1% at 0.4–1.8σ of the mean (`data_release/ensemble_2026-09-26.tsv`, `ensemble_1p_2026-09-26.tsv`).

15.4 Defects found during the adoption

- **Ensemble throws on NFS.** Writing each throw to a temporary name and renaming it over the final one (2026-09-24) did not stop the race: on NFS the rename unlinks the old file under a reader on another node, which then fails with a short read (two FHC p_μ members within two minutes of five arrays starting together). The arrays were cancelled after six minutes. A member's throw is now created once by `link(2)` from a closed temporary, stamped with its seed and POT scale, reused when the stamp matches and never replaced while it may be open (`macros/throw_guard.h`, `ens/throws/`). The FHC and RHC arrays of one observable also shared a work directory, a universe log and a file list per member; these are now per configuration.
- **Response figure of p_π.** The figure script skipped the p_π response, so the note showed the two-region response of an earlier release; it now writes `fw_resp_ppi3bin` and `fw_eff_ppi3bin`.
- **MCS tool.** It read the MCS extractions of the retired binnings for $\cos\theta_\mu$; it now checks whether the reconstructed bins read the muon momentum and writes the identically zero term otherwise.
- **Proton kinetic energy.** The proton candidate's kinetic energy is the range-based value, an exact function of the track length, not a calorimetric one; the notes said otherwise.
- **Reproduction precision.** The external reproduction of the released curves was quoted at 1.2×10^{-10}; from the released text files it is a relative 4×10^{-8}, the precision of those files.
- **Hand-maintained tables.** The note's table of bin edges and the supplement's tables of A_C row sums and proton-tagged binnings carried values of earlier releases; the first two are now generated (`binning_table.py`, `ac_rowsums_table.py`).
- **Combined-truth test.** The check of the combined truth against the average of its inputs used POT weights and A_C-smeared integrals; with fluence weights on the one-bin truths the ratio is 1.000 (inclusive) and 1.001 (proton-tagged).
- **Top bin closed at a physical boundary.** The last true bin of $\cos\theta_\mu$ and $\cos\theta_\pi$ is written as "< 1.000" (likewise $\delta\phi_T$ at 180° and $\theta_{\pi p}$ at 3.142 rad), so an event exactly at the boundary falls in no true bin. Five of the 145 946 signal events of the fourteen overlays have $\cos\theta_\mu = 1.0$ exactly (two of them selected), none has $\cos\theta_\pi = 1.0$, and no ensemble member is short by more than this. The effect is 0.003% of the signal and is left in the release; the configuration writer now leaves such a bin open at the top for the next rebuild, and the ensemble entry-count scan (`report/tools/ensemble_entry_scan.py`) reports a shortfall of one or two entries as this case rather than as a truncated read. The one-entry-short members flagged on 2026-09-24 were this case, not corrupt reads.

16 Response to the collaboration review (2026-09-27)

16.1 Decisions

- Four status labels only: *Primary result*, *Secondary result proposed for approval*, *Validation-only product*, *Not reported*. The status of every product is kept in one file,

`report/status.tsv`, from which the status tables of both notes and the status column of `data_release/index_extractions.tsv` are generated.

- Proposed for approval as secondary results: the proton-tagged one-bin total; the proton-tagged θ_p and $\theta_{\pi p}$ (FHC, RHC, combined); the two-dimensional $\theta_p \times \delta p_T$ (proton-tagged sample) and $\cos\theta_\pi \times \cos\theta_\mu$ (inclusive sample), combined configuration only. Earlier drafts called θ_p, $\theta_{\pi p}$ and the two pairs "proposed secondary" in some places and validation-only in others. The inclusive pair moved from the proton-tagged note to the analysis note.
- The MCS muon-momentum-scale term is part of the official covariance of every result (`cov_total_plusMCS.txt`); every uncertainty and model-comparison table is regenerated with it (`report/tools/official_cov.py`). FHC p_μ moves from 39.3% to 40.6% systematic and from 41.2% to 42.4% total; the model χ^2 of p_μ decrease (FHC GENIE 2.73 $\rightarrow$ 2.24). The closure χ^2 and the robustness biases stay with the framework covariance.
- The independent-generator closure is defined: a validated NuWro overlay, every bin holding at least 10% of the largest truth within 1σ, every integral within its uncertainty (U3).

16.2 Release

- θ_p and $\theta_{\pi p}$ (three configurations) and the two combined two-dimensional results were added to the data release with their covariances, A_C and curves (`xsec_analyzer/slurm/harvest_proposed_local.sh`, `report/tools/export_2d.py`). The re-harvest reproduced the unfolded results, truths and A_C exactly; the combined generator curves of the two-dimensional results moved by 0.3–1.1%, because their files predated the correction of the combined generator weights ($8.857 \rightarrow 7.766 \times 10^{20}$ FHC exposure).
- The MCS term of the new entries is exactly zero for θ_p, $\theta_{\pi p}$ and $\cos\theta_\pi \times \cos\theta_\mu$; for $\theta_p \times \delta p_T$, which uses the muon momentum through δp_T, it was extracted (SLURM 3433847): at most 16% of a cell and 0.20σ, integral unchanged.
- The release README states the 60 MeV flux convention, the status labels, the official covariance and the file schema.

16.3 Documents

- Both notes open with the same decision box and a status table; the inclusive note has a measurement-definition table, a section on the migration criterion and binning rule, the validation ladder, and a numbered prerequisite table with criteria, owners, evidence and blocking levels, shared verbatim by the three documents (`report/shared/`).
- The proton-tagged note is restructured: decision and status, scientific purpose, selection, observables, extraction, validation, proposed results, validation-only products, products not reported, prerequisites. Reconstruction validity (P4) and model sensitivity (P3) are separate prerequisites.
- The figures use physics notation, the open-ended last bins are hatched and labelled, and the six-panel proton-tagged figures are split in two.
- The proton-tagged W_{had} and TKI observables had no generator predictions in RHC and combined, and the muon and pion observables of the proton-tagged sample none in any configuration. The RHC and combined W_{had} and TKI predictions existed but were not referenced by the configurations; the muon and pion ones were produced with new readers that reuse the proton-tagged signal definition (`generator_predictions/gen2d/*_1p_inc*`, `regen_1pinc.sh`). The 25 extractions were unfolded again: unfolded results, truths, A_C and every covariance are bit-identical, and only the generator curves were added.

- Terminology: pseudo-data, control region, combined configuration (`report/tools/terminology.py`). A cross-document check, `report/tools/check_documents.py`, enforces the shared sources, the status vocabulary and the adopted exposures, fluxes and target count.

17 Muon momentum-scale term for the proton-tagged observables (2026-09-27)

The data-side MCS momentum scale ($\pm 5\%$) is evaluated for the proton-tagged observables that use the muon momentum: p_μ, W_{had}, δp_T, $\delta\alpha_T$ and p_n, in FHC, RHC and combined (SLURM 3433817, `xsec_analyzer/run_mcs_1p.sh`). The scaled fake data (`macros/mcs_scale_fakedata_1p.C`) recompute W_{had} and the TKI variables from the scaled muon; with a scale of one they reproduce the stored branches exactly. The largest term is 0.62σ (FHC, open highest p_μ bin); for the derived observables it is at most 0.37σ (lower W_{had} region, combined). $\delta\phi_T$ and the observables that do not use the muon momentum carry exactly zero. Released as `cov_MCSscale.txt` and `cov_total_plusMCS.txt` under `data_release/cov/1p_*` and summarised in `data_release/mcs_scale_1p_2026-09-27.tsv` (`report/tools/mcs_eval.py all 1p`); not yet inside the framework covariance.

18 Document revision of 2026-09-27

The three documents were revised for order and consistency. No result moved; the items below are statements that were wrong or inconsistent and are now corrected, and material that moved between documents.

18.1 Beam-on control-region material held out

The three documents are written as they stand before the control regions are compared with beam-on data. The eighteen passages that used beam-on data (the comparison subsection of the analysis note with its figure, and the sentences of the abstract, introduction, limitations, unblinding requirements and summary, one clause of the proton-tagged note, and the control-region outcome and one cross-check clause of the supplement) are kept verbatim, with their replacements, in `report/beam_on_held_out.tex` and `beam_on_held_out.json` for merging back. The analysis note label `sec:cr_outcome` is replaced by `sec:cr_procedure`.

18.2 Inclusive analysis note

- The RHC paragraph of the results stated that no RHC dirt sample exists and that the FHC dirt overlay was scaled to the RHC exposure. The RHC configuration uses the RHC dirt overlays of Runs 3b, 4a and 4b, with Run 3b standing in for Runs 1 and 2; the text and the sample table now say so.
- The summary quoted 30–46% bin-averaged uncertainty without data statistics and 67% for FHC p_π; the tables give 33–46% and 57%. The column "Total" of the per-configuration result tables is the systematic total without the data statistics and is now labelled so.
- The detector appendix stated that track momentum is obtained from curvature (there is no magnetic field), that the NuMI spill is 1.6 μs (that is the Booster spill; NuMI is about 10 μs), that the signals are digitised inside the cryostat, and listed a cosmic BDT that the selection does not use. It was rewritten. The proton-tagged note no longer repeats it.

- Proton-tagged numbers were removed from the inclusive note (the proton-tagged column of the systematics table, the proton-tagged rows of the total cross-section table, the proton-tagged binning, cut parameters, alternative binnings and $W_{\pi p}$ appendix) and now appear only in the proton-tagged note, which has its own totals, binning and inventory tables. The generators (`rebuild_tables.py`, `binning_table.py`, `inventory.py`) write one table per note.
- The validation of the background fit (toys, injections, Markov chain, pull-mean convergence) moved to the control-region part of the technical supplement.
- Release dates were removed from the running text of the ensembles and appendices.

18.3 Proton-tagged note

- Restructured in the order introduction, selection, observables and binning, extraction and uncertainties, results, validation, prerequisites.
- The text on the pair $\theta_{\pi p} \times W_{\pi p}$ quoted a conditioning of 0.560, a 58% data-statistical term, per-bin uncertainties up to 545% and a closure ratio of -0.40, values of an earlier extraction; it now quotes the table (0.537, 78.3%, up to 276%, down to -0.80).
- The efficiency is quoted as 11.5% (FHC), as in the cut-and-count table, not 11.4%; the largest detector term as 75%, not 100%.

18.4 Technical supplement

- Reordered into parts on pion momentum and binning, proton-tagged observables, extraction and validation, control regions, and selection and infrastructure: the pion-estimator survey and the five-bin p_π response moved to the pion-momentum section, the D'Agostini cross-check, the total cross section, the A_C row sums and the ensembles to the validation part.
- The abstract stated that p_π is reduced to two bins; it is measured in three regions.
- A reference intended for the extraction ensembles pointed to the background-fit toys.
- Prose revised throughout; tables, figures and numbers unchanged.

19 Validation of the results proposed for approval (2026-09-28)

Prerequisite V1 is complete. The results proposed for approval on 2026-09-27 that had not been through the validation of the other extractions, θ_p and $\theta_{\pi p}$ (FHC, RHC, combined) and the two combined-configuration two-dimensional results, $\theta_p \times \delta p_T$ and $\cos\theta_\pi \times \cos\theta_\mu$, now have both tests.

- **Alternative models.** GENIE multisim universe 545 and the $\Delta \to N\pi$ angular unisim, as for the other extractions: the proton angles in FHC and $\cos\theta_\pi \times \cos\theta_\mu$ (SLURM 3433849), $\theta_p \times \delta p_T$ (SLURM 3433979), in the combined configuration for the two-dimensional results (`xsec_analyzer/run_altmodel_v1.sh`, `configs/alt`, `report/tools/altmodel_eval.py V1`). No bin moves by more than 0.79σ, the second θ_p bin under the GENIE variation, which returns all four θ_p bins 9–28% low while their truth moves by -3%; the two-dimensional results stay within 0.74σ (proton-tagged) and 0.61σ (inclusive).
- **Ensembles.** 100 members for each of the eight extractions (SLURM 3433855–3433862, `xsec_analyzer/ens_v1.sh`): statistical pull widths 0.94–1.04, coverage nominal, no truncated read in the 4800 inputs (33 inputs of the inclusive two-dimensional ensemble are one entry short, the boundary case of an event at $\cos\theta_\mu = 1$ exactly). In the sparsest inclusive

two-dimensional cell ($\cos\theta_\mu < 0.65$, $\cos\theta_\pi < 0.35$) the ensemble mean lies 11% below the reference, 0.58 statistical and 0.17 total standard deviations, and the ensemble-mean integral lies 1.2% below it (3.4 standard errors of the mean); no correction is applied (`data_release/ensemble_v1_2026-09-28.tsv`).

- **Status.** V1 is removed from the prerequisites of the three products in `report/status.tsv`, and the prerequisite table marks it complete.
- **Tools.** The statistical-covariance step (`slurm/ens_statcov.sh`) and the pull evaluation (`report/tools/ensemble_stat_pulls.py`) accept two-dimensional extractions, whose pulls are read from the all-slice closure file. The proton-tagged ensemble table has one row per extraction and includes the new ones (`report/tools/ensemble_tables_v1.py`, `ensemble_tables_1p.py`).

19.1 Documents

- The authors' revision of the wording of the three documents is adopted; no number changed.
- The label "Secondary result proposed for approval" was made plural by appending "s", which printed "... proposed for approvals"; a plural form is now defined in `shared/status_vocab.tex`.
- Four tables that ran into the margin were narrowed.

20 Approval material moved to a separate document (2026-09-28)

The analysis note, the proton-tagged note and the technical supplement describe the analysis as it stands. The material of the approval process moved to a new document, `approval_request.tex`:

- the decision requested from the collaboration (no longer boxed), the status tables with their prerequisites column, the prerequisite table with the criteria of U1–U7, P1–P4 and R1, and the validation ladder;
- from the supplement, the request to inspect the control regions, the frozen analysis state, the points the collaboration is asked to ratify, the approval checklist, the approval rules within the frozen procedure and the sign-off.

In the three documents the validation ladder is replaced by paragraphs of text, the sections on prerequisites by sections on limitations, and statements of what is still needed by statements of what has and has not been done. The analysis note starts with the introduction, followed by the measurement definition; its status table is in the results section. The status tables of the three documents no longer carry a prerequisites column (`report/tools/inventory.py` writes both versions; the closing quotes of "yes" in their captions were lost to a quoting error and are restored). The three boxed statements of the supplement are text. `report/tools/check_documents.py` now refuses a prerequisite identifier in an analysis document and accepts the decision list, the prerequisites and the validation ladder only in `approval_request.tex`. The status label "Secondary result proposed for approval" is shortened to "Secondary result" in the documents, the release index and the checks. No number changed.

21 Collaboration review of 2026-09-28: corrections and shorter documents

The review was answered without adding material; the three documents are shorter (analysis note 63 → 61, proton-tagged note 22 → 21, supplement 78 → 56 pages). No result changed.

- **Corrections.** The proton-tagged flux arithmetic now states that 298/331 is the flux-dependent neutrino background over the signal of the total-cross-section table (whose other 7 background events are beam-off and dirt), and that the propagated universes give the remaining 1.9% of the 35.5%. The proton-tagged combined integrals lie between FHC and RHC for nine of eleven extractions (not ten of twelve), and span 0.38–0.57.
- **Wording.** The abstract of the analysis note puts the pseudo-data status first and no longer says "we measure"; the reweighting test is called a test of reweighted interaction-model variations, and the limitations state that it cannot change final-state multiplicities; the proton-tagged total is described as the cross section of the final states with a proton above 0.3 GeV/c, and the generator spread as about one standard deviation of the measured total; "response matrix" replaces "smearceptance" in the prose. Total-cross-section captions state that the 60 MeV flux includes the flux below the pion-production threshold; validation captions state that biases and full pulls use the framework covariance without the MCS term; the four statuses are defined in one sentence (`\statDefinitions`).
- **Text extraction.** Every preamble maps glyphs, ligatures included, to Unicode (`glyphtounicode`), so copied text no longer contains ligature characters.
- **Shorter.** Supplement: the physics-background part removed (the analysis note keeps its physics context), the rejected alternatives condensed to one record each, the control-region machinery condensed (the stale FHC yields of the former uncertainty table and of the π^0-cut study, open item D1, went with them). Analysis note: the detector appendix in two paragraphs, the repeated limitation paragraphs and eight uncited references removed. Proton-tagged note: one results table for the three configurations (`rebuild_tables.py`, `check_tables.py`).

22 Version 1.6 for the Editorial Board (2026-09-29)

The authors' revision of the three documents is adopted and tagged as version 1.6 (git tag `v1.6`). The beam-off normalisation is described by trigger rather than gate counts; the status tables, the status labels of the proton-tagged note, the documents diagram, the glossaries of the notes and the subsection on the comparison with beam-on data are removed; the analysis note states the publication strategy that the separation into documents reflects. The supplement has its own preamble; a duplicated block of cross-reference commands in it was removed, and references to the removed tables, appendix and subsection were redirected or dropped. No number changed.

23 Dirt normalisation in the count-level tables (2026-10-01)

The cut-flow and cut-and-count macros (`cutflow_yields.C`, `cutflow_yields_1p.C`, `total_xsec_counting.C`) scaled the Run-1 dirt sample by the data POT over the summed overlay POT (0.081020 FHC, 0.071666 RHC), a placeholder introduced on 2026-08-09, instead of over the POT of the dirt samples; the last two also applied the 0.65 dirt weight twice. The macros now scale each run's dirt sample by the data POT over its own POT, as the extraction does. The extraction is unchanged: the released one-bin FHC total already subtracts a background of 590.0 events, 6.6 of them dirt. The dirt at the final cut becomes 6.6, 9.5 and 16.0 events (FHC, RHC, combined) instead of 1.1, 1.0 and 2.1; the purities become 58.5, 57.4 and 57.9% (51.9, 49.7 and 50.7% proton-tagged). The cut-flow tables, the cut-and-count tables, the cut-flow figures and the quoted purities are regenerated. The control-region macros carried the same placeholder; their correction follows.

24 Dirt normalisation in the control regions (2026-10-01)

The control-region macros (`sideband_compare.C`, `sb_protocol.C`, `sb_protocol_data.C`, `sb_plots.C`, `sb_plots_data.C`, `sb_perrun_data.C`, `bkgfit_templates.C`) and three scripts found with them (`bkgfit/farsb.py`, `bkgfit/run1_trigger.py`, `pi0final/sb_pi0_final_yields.C`) used the same placeholder, the 0.65 applied twice in most of them. All now take each run's dirt sample scaled by the data POT over its own POT, from the file list of the release (`macros/dirt_perrun.h`, `bkgfit/dirt.py`). The control-region dirt rises seven- to twentyfold: 342/460/803 events in CC0π (FHC/RHC/combined), 93/124/216 in π^0, 18.5/35.8/54 in the cosmic region and 0.5/1.7/2.1 in multi-π. The neutrino and beam-off components, the selection and the cross sections are unchanged. In the analysis note the dirt column of the control-region yields, the purity of the cosmic region (4.1%) and of the selection without the opening-angle cut (54.2%), the beam-off share of the cosmic region (33/26%, now of the complete prediction) and its statistics (283 events, 21%) change. In the supplement the full-stack test (0.96–1.12), the frozen-procedure statistics on pseudo-data (global $\chi^2 = 2.29/8$, pulls within ± 0.8, shape tests $p > 0.12$, single-region power 12% and 44–62%, common-shift blind spot 50%, null calibration 5.2% and 1.1%), the shares of the fixed components, now with the dirt, and the background-fit validation change. These numbers, the proton-tagged control-region table and the beam-off shares had not been regenerated after the Run-1 exposure correction of 2026-09-19 and now include it. With beam-on data (control-sample note) the frozen global test gives $\chi^2 = 6.03/8$ ($p = 0.64$; it was 8.11/8), the joint fit gives CC0π at 0.975 ± 0.191, the FHC Run-1 residual is unchanged (0.862 ± 0.021), the cosmic region moves from 1.21 to 0.98 and its far versions fall to 0.79 and 0.76. The supplement's comparison with the ν_e analysis listed a 100% dirt normalisation uncertainty among the shared conventions; this analysis carries none, and the comparison now says so, with the bound that a 100% variation moves the one-bin totals by at most 0.9%.

25 Version 1.6.1 for the Editorial Board (2026-10-01)

The three documents with the two dirt corrections above are tagged as version 1.6.1 (git tag `v1.6.1`), an errata release of version 1.6. The cross sections, the covariances and the data release are those of version 1.6; the release notes list every changed number.

26 Bragg-pion requirement dropped from the pion identification (2026-10-02)

The pion identification of the single-pion selections required a Bragg-pion score of at least 0.08. The beam-on, beam-off and dirt ntuples of the older production (FHC Run 1, RHC Runs 1 and 3b) do not carry this score, so the requirement passed every track in those samples while every overlay applied it. In the opened control regions the released identification passed 7–11% more beam-on candidates than predicted in those three periods and not in the others; without the requirement the periods agree, and where the data carry the score the requirement, after the other cuts, is modelled to within 2.2σ. At unblinding the one-bin cross sections would have been 4.7% (combined, inclusive) and 4.6% (proton-tagged) high. The requirement is dropped in every sample (commit `1fb6be9`); the pion BDT, LLR and MIP-BDT requirements are unchanged.

The selection keeps 18.3% of the signal at 57.3% purity in FHC (18.4% and 56.6% in RHC, 18.4% and 56.9% combined; previously 17.5–17.6% at 57.4–58.5%), and the proton-tagged selection 12.1% at 50–52% (previously 11.5%). The pion candidate is a proton in 21% of the selected events (18%), and the CC0π background rises from 215 to 260 events in FHC, 42% of the neutrino background. The one-bin totals become 1.042, 0.925 and 0.975 (FHC, RHC,

combined; previously 1.034, 0.932, 0.975), the differential integrals move by factors of 0.92–1.12 (median 1.00), and the mean inclusive closure ratio is 1.003.

Every overlay, beam-off, dirt and detector-variation sample of both trees was reprocessed, the pseudo-data were re-thrown with the release seeds (the same events; only the selection flags change), the 65 universe files were rebuilt with the bin definitions read back from the previous files, and the release chain, the D'Agostini cross-check, the cut-flows, the tables and the figures were regenerated. The proton-rejection figure now shows the pion-BDT score. The final-cut topology figure, which still showed a NuWro pseudo-data set at the superseded Run-1 exposure, is redrawn from the release samples. The counting cross-check now takes its systematic fractions from the current p_μ results, and the proton-tagged cut-flow dirt carries the proton-tagged counters (it had been processed with the inclusive selection). The binnings still meet the 0.50 migration criterion (lowest diagonal 0.516, the inclusive three-region p_π in FHC).

Carried over from version 1.6.1, made with the previous selection: the study extractions and the binning-review studies, which remain the dated basis of the binnings. The MCS momentum-scale term, the alternative-model closure and the ensembles are redone (following sections). Dropping the requirement moves events from the control regions into the signal region: about 6–7% of the new signal region of the Run-4 and Run-5 beam-on data (about 85 predicted events) was in the opened control regions. The beam-on control-region skims made with the previous selection are withdrawn from use and have been remade; in Runs 1–3 the beam-on selection does not change.

27 Muon momentum-scale term and alternative-model closure with the new selection (2026-10-03)

The data-side MCS term was evaluated again on pseudo-data re-thrown from the reprocessed samples with the release seeds. In p_μ it is at most 0.60σ (FHC, open highest bin; previously 0.56σ), 0.44σ combined and 0.32σ in RHC; for FHC p_μ it is 9.7% bin-averaged (previously 8.2%) and the total uncertainty 41.4%. For the proton-tagged observables it is at most 0.66σ (p_μ, FHC), and 0.38σ for the derived observables. The official covariances, the summary tables and the systematic fractions of the counting cross-check are regenerated. The statement that the downward scale leaves the two-bin δp_T unchanged no longer holds and is removed.

The alternative-model closure was repeated with the same universes and seeds. The largest inclusive bias is 1.10σ, in the third FHC $\cos\theta_\mu$ bin under the multisim variation (previously 1.18σ in the same bin under the angular variation), and RHC stays within 0.85σ. The low-imbalance proton-tagged δp_T and p_n bins move by -0.99σ and -0.97σ under the multisim variation (previously -1.14σ and -1.16σ); the proton angles stay within 0.79σ, and the two-dimensional results within 0.74σ (proton-tagged) and 0.55σ (inclusive). The note stated that the third FHC $\cos\theta_\mu$ bin has the smallest A_C row sum of the eight; the second bin has, in version 1.6.1 as now, and the statement is removed.

The truth-level changes of the two alternative models are now the ratio of the varied to the central-value universe in the release universe files. They had been read from the truths of two independent throws, which adds several percent of Poisson noise per bin. Under the multisim variation the signal rate changes by 3.5% (FHC) and 2.5% (RHC) and the released $\cos\theta_\pi$ and p_π bins by up to 30% and 22% (previously quoted as 4%, 29% and 24%); the RHC p_μ truth rises by 2–3% (previously 3–9%), the near-threshold FHC p_π truth falls by 18–20%, the true cells of $\cos\theta_\pi \times \cos\theta_\mu$ change by -23% to $+16\%$, and the proton-tagged low δp_T and p_n truths rise by 2%.

28 Report folder organised by kind (2026-10-04)

The analysis documents, their shared inputs (`shared/`, `glossary.tex`, `common_preamble.tex`, `status.tsv`) and the control-region note are in `report/notes/`; papers, slides, other notes, plans, result tables, figure macros, release packages and earlier exports are in `papers/`, `slides/`, `other_notes/`, `planning/`, `results/`, `macros/`, `releases/` and `archive/`, and `report/README.md` lists them. Figures, tables, the data release and the tools are where they were. Every release table was regenerated in the new layout: the inclusive and proton-tagged p_π region tables of the supplement had not been regenerated for version 1.7 and now carry its values, as the corresponding figures did.

29 Ensembles repeated with the new selection (2026-10-05)

All seventeen ensembles of the release, seven inclusive and ten proton-tagged with 100 members each, were rebuilt from new throws of the reprocessed samples. The statistical pull widths are 0.92–1.03 (inclusive) and 0.93–1.10 (proton-tagged) with nominal coverage (previously 0.95–1.07 and 0.94–1.08). The one-dimensional ensemble means lie within 0.7% of the reference at up to 1.5σ on the mean (previously 0.4–1.1% below at up to 2.0σ); that of $\cos\theta_\pi \times \cos\theta_\mu$ lies 0.9% below at 2.7 standard errors, carried by its sparsest cell, which lies 0.55 statistical standard deviations low on average (0.58 before). Of the 9200 inputs, 9129 were read in full and 71 miss one or two events at the edge of the true binning; no member read an incomplete file. The summary files `data_release/ensemble_2026-10-05.tsv`, `ensemble_1p_2026-10-05.tsv` and `ensemble_v1_2026-10-05.tsv` replace those of 2026-09-26 and 2026-09-28.

30 Figures, control regions and wording checked against version 1.6 (2026-10-05)

Every figure was checked against its source and every control-region number against the control-region chain of version 1.7. Regenerated: the two-dimensional cross-section figures, the proton-tagged cut-flow figures (the release script built them without copying them into the report; fixed), the new-observable figure, the resolution figures, the per-run beam-off figure with `data_release/ext_gates.tsv`, and the null calibration of the control-region test. Updated from the version-1.7 chain: the control-region yields and pseudo-data test of the analysis note; in the supplement the region roles, the full-stack test, the transfer-factor uncertainties and their floors, the frozen-procedure statistics (global χ^2 1.49/8 on the pseudo-data, null rate 5.1%), the CC0π shares of the components held fixed, the proton-tagged control regions and the validation of the background fit. Also updated: the beam-off rates per run period, the resolution captions, and the closure, conditioning and uncertainty figures of the proton angles and of the two two-dimensional results. The text otherwise follows version 1.6: wording changed beyond the numbers in the passes above is restored, except two statements that no longer hold, the A_C row-sum clause on the third FHC $\cos\theta_\mu$ bin and the statement that the downward muon-momentum scale leaves the two-bin δp_T unchanged.

31 Version 1.7 (2026-10-03)

The three documents with the changes above are version 1.7. The cross sections, covariances and data release are regenerated; the release notes list the changed numbers.

32 Detector variations taken fractionally (2026-10-08)

The Run-4 detector-variation samples were simulated with the earlier NuMI flux and the nominal overlays with the current one (Geant v4.10.4). Per POT the variation CV samples predict 1.27 (FHC) and 1.51 (RHC) times the nominal event rate in the fiducial volume, the ratio rising from 1.1 below 0.5 GeV to 1.8–3 above 5 GeV; weighted event by event with the ratio of the two fluxes in true energy and angle to the beamline, they predict 0.94 times the nominal rate in both modes, flat in energy. The detector covariance was built from the absolute difference between each variation and the variation CV, scaled to the data POT, and so carried that rate ratio. It now applies the fractional change of each variation relative to the variation CV to the nominal prediction (covariance type `DVfrac`, commit `30275ee`). Weighting the variation samples to the current flux as well changes the fractional detector change of the selected count by at most 0.3% absolute (8.2 → 8.4% FHC, 8.8 → 9.1% RHC) and is not applied.

Per bin the detector uncertainty falls by a median factor 0.78 (FHC), 0.70 (RHC) and 0.71 (combined), and the total uncertainty by 0.94. The one-bin totals keep their central values; their detector term is 14.0–14.3% inclusive (was 19.2–22.1%) and 8.6–16.8% proton-tagged (was 11.7–26.7%), their total uncertainty 34.1–35.9% (was 37.7–38.2%) and 40.5–42.9% (was 41.4–47.6%). The Wiener filter is built from the covariance, so the unfolded values and A_C of the differential results change as well: their integrals move by a median factor 1.026 (0.92–1.18), and A_C retains more shape in most results (inclusive combined $\cos\theta_\mu$: trace 3.81 → 4.23). The control regions are unchanged; their detector model was already fractional.

33 Version 1.8 (2026-10-09)

The three documents with the change above are version 1.8. The cross sections, covariances, data release, MCS term and alternative-model closure are regenerated, and the members of the seventeen ensembles of version 1.7 are unfolded again on their existing universes; the release notes list the changed numbers. The fractional treatment is described where the covariance is defined (analysis note, systematic uncertainties and the RHC samples; supplement, covariance). Four supplement items that still carried values of earlier versions now carry those of version 1.8: the D'Agostini table, the mean A_C row sums of the proton-tagged observables, the passages on the p_π regions and the per-observable ratios of the combined truth to its inputs. Changed because they no longer hold: no released A_C has a negative row (the top row of RHC $W_{\pi p}$ sums to 0.24, was −0.03); the RHC $\cos\theta_\mu$ ensemble mean now lies below the reference like the others; FHC $\delta\phi_T$ is no longer among the proton-tagged results with the largest detector term; the identity check now has one non-$\cos\theta_\mu$ extraction above 4×10^{-6}, so the quoted threshold is 6×10^{-6}.

34 MCS calibration of exiting muons (2026-10-10)

In the opened control regions the MCS momentum of exiting muon candidates is lower in the beam-on data than in the simulation: above 1.2 GeV/c the data hold half the predicted share of exiting candidates, and the released p_μ bins of the control-region events reach 0.48 of the predicted shape in the highest bin. The MCS and range momenta of contained muons agree between data and simulation within 1.5% up to 3 GeV/c, long tracks are as straight in the data as in the simulation, and among the detector variations only the space-charge variation moves the MCS momentum, by about a fifth of the difference. The data carry an angular noise on the scale of the MCS segments that the simulation lacks; the effect is strongest for tracks with a large drift component and for short tracks. The studies are in the control-sample note. Left uncorrected, pseudo-data with the MCS momenta of the data would move the released p_μ bins by up to 2.7σ and their integral by 25–29%, against the ±5% momentum-scale term of 0.65σ at most.

From version 1.9 the MCS momentum of simulated exiting muon candidates is mapped to the control-region data in twelve cells of track direction and length (quantile maps; beam-off events and overlaid cosmic tracks are not mapped), in every simulated sample, and the universes of the results that use the muon momentum are rebuilt: p_μ (inclusive and proton-tagged), W_{had}, δp_T, $\delta\alpha_T$, p_n and $\theta_p \times \delta p_T$, nineteen results. The uncertainty of the calibration (seven alternative maps in four fully correlated terms) replaces the ±5% momentum-scale term in the official covariance (`cov_MCScal.txt` for `cov_MCSscale.txt`). With the calibrated response two binnings no longer meet the 0.50 migration criterion and are changed: the two highest p_μ bins (column-normalised diagonals 0.43–0.49 and 0.24–0.30) are merged into one bin above 0.825 GeV/c (0.61–0.69), and the θ_p edge of $\theta_p \times \delta p_T$ moves from 0.8 to 0.85 rad (third cell 0.488 → 0.506; chosen as the admissible binning closest to the released one among 143 screened). The generator predictions of both are regenerated. The other forty results are unchanged, bit for bit.

35 Version 1.9 (2026-10-10)

The three documents with the change above are version 1.9. The nineteen results, their covariances, the data release, the alternative-model closure and the five ensembles of these results (FHC p_μ, FHC W_{had} and δp_T, combined p_n and $\theta_p \times \delta p_T$) are regenerated; the ensemble members are rebuilt from calibrated copies of their throws. The calibration is described where the muon momentum and the MCS term are defined (analysis note, selection and the MCS subsection; proton-tagged note, MCS subsection; supplement, covariance), and the binning appendix and the proton-tagged binning record the two binning changes. The release notes list the changed numbers. The nineteen results close with $\chi^2 \leq 2.41$ ($\theta_p \times \delta p_T$, four cells); the inclusive p_μ integral ratios, 0.905 (RHC) to 1.106 (FHC), lie within twice the throw-to-throw spread of the FHC p_μ ensemble (6.2%), whose mean is 0.7% below the reference. Changed because they no longer hold: the largest proton-tagged detector term (55%; the quoted 75% was a value of version 1.7), the proton-tagged W_{had}, which now meets only the 0.50 criterion, and the statement that the comparison of the control regions with data is unaffected by the MCS treatment.

36 Control regions with the calibrated simulation (2026-10-11)

The frozen protocol was repeated with the calibrated overlays, dirt and detector variations of version 1.9 (option `SB_MC_DIR` of the protocol macros; beam-off and beam-on data unchanged). Every shape test passes: the four p_μ tests that failed (CC0π and π^0 in both modes) fall from $\chi^2 = 13.3$–66.2 to 0.8–3.8 for 6 degrees of freedom, and the sum over the eight p_μ tests from 189.5/48 to 20.5/48 (22.9/48 for the pseudo-data of the calibrated simulation). The region totals, pulls, global test and the other shape tests do not depend on the muon momentum and are unchanged. The map was derived from the same events, so the comparison is a closure per region and beam mode. The control-sample note records it; the results are unchanged.

37 Hold-out test of the MCS calibration (2026-10-16)

The nominal maps were derived on part of the control-region candidates and tested on the rest (`scripts/mcs_cal_holdout.py`). With random halves the p_μ bins of the test half agree with the data shape within 6% ($\chi^2 = 21.1$ and 5.9 for 5 degrees of freedom, against 7.5 ± 4.0 from statistics alone). Maps from FHC tested on RHC and the other way round, and from the old production tested on the new and the other way round, move the highest bin by 3–7%, as much as the maps from period groups that make the beam and production terms of the calibration uncertainty. The results are unchanged.

38 Open items

The following remain outstanding following the 2026-09-07 review. None is a correction awaiting application; they are prerequisites, limits of the exposure, or tests not yet made. The analysis note's request checklist carries the gating items; its Limitations section carries the tests not yet made.

- **Before the control regions are opened:** ratification of the frozen procedure by the conveners, including the π^0 definition and target class (adopted by the analysers 2026-09-06) in the light of the multiplicity-cut comparison now available; independent verification of the control-region skim (implemented 2026-09-06) by a second person, and the hashes of the frozen skim, plotting list and protocol macro in the approval record (repository tag `cr-freeze-2026-09-06`); the decision whether the π^0 region is diagnostic-only or eligible for a fitted correction, taken after the pre-fit result.
- **Before the signal region is opened:** confirmation that the flux constants are the PPFX-corrected central values; a NuWro or GiBUU fake-data pass through the released configurations (no alternative-generator test exists on the released chain); the clean-cache bit-identical reproduction repeated on the 2026-09-06 release; the inverted-proton-PID control region for any proton-tagged result; the documented treatment of any control-region discrepancy under the frozen procedure.
- **Ensembles:** every released extraction with an ensemble was repeated for version 1.7 (2026-10-05) and unfolded again for version 1.8 (2026-10-09): statistical pull widths 0.92–1.09, coverage nominal, integral offsets within 1.1%. No test of this kind remains outstanding.
- **Systematic terms not in the released covariance:** the MCS-calibration term is evaluated on the data side and released as a separate component for the nineteen results that use the muon momentum (for p_μ up to 0.28σ per bin in FHC, 0.43σ combined and 0.21σ in RHC), is exactly zero for the other results, and is not propagated inside the framework; the beam-off gate-ratio and Run-2 stand-in terms ($\lesssim 0.4\%$ of the total); the beamline-geometry flux term ($\lesssim 0.2\%$).
- **Muon-momentum shape in the control regions:** the two-dimensional comparison with beam-on data shows a residual organised in p_μ (data over prediction 1.16–1.82 at 0.35–0.75 GeV/c) in both modes and in regions of different composition. It is the MCS momentum of exiting muons, calibrated from version 1.9; with the calibrated simulation the released p_μ bins of the control-region events agree with the data shape within 4%, and every shape test of the frozen protocol passes. Maps derived on half of the events describe the other half within 6%, close to the statistical expectation, and maps derived on one beam mode or production describe the other within the spread of the maps from period groups alone (control-sample note).
- **FHC Run 1 residual:** the $13 \pm 2\%$ offset of FHC Run 1 relative to the other five periods in the control regions, which survives the exposure correction and is not covered by any correlated term; its cause (bookkeeping POT, completeness of the Run-1 beam-on sample against its run list, or the Run-1 overlay normalisation) is not identified.
- **Limits of the exposure:** the differential $W_{\pi p}$ shape is withdrawn (no binning satisfies both the migration and the statistics requirement); the proton-tagged sample supports an integrated cross section.
- **Ensemble offset:** the inclusive one-dimensional ensemble means sit within 0.9% of the reference, all six below it, at up to 1.5σ (the RHC total, -0.6%), and that of $\cos\theta_\pi \times \cos\theta_\mu$ 0.8% below at 2.4σ, carried by its sparsest cell; the throw itself is unbiased (mean thrown

selected count 1.004 of the expectation), so the offset, if real, is in the extraction; a larger ensemble would settle it.

External reproduction of the A_C-smeared comparison (resolved). The transformation

$$(d\sigma/dx)_i^{\text{smeared}} = \frac{1}{w_i} \sum_j (A_C)_{ij}\, p_j,$$

applied to a prediction supplied as a per-bin cross section, reproduces every released generator curve to a worst relative per-bin deviation of 4×10^{-8}, the precision of the text files; it was the failure of this test that exposed the double-A_C bug, and its success is now the regression check. No packaged reference implementation ships with the release; a user can verify their implementation against any model column of `curves_*.tsv`.

A caveat that is not an open item but reads like one. The closure χ^2 values have p-values clustered near unity (median 0.98 over the fifty-one extractions) because the measurement and the truth it is compared against share their systematics, so those terms cancel in the difference while still inflating the covariance used to judge it. The ensemble of throws sizes the statistical part of the band correctly; the systematic part remains untested by construction.

39 Superseded values that may still be encountered

Quantity	superseded value	status
Mean inclusive closure ratio	1.106, 1.001, 1.016, 1.018, 1.010, 0.996	1.001 (version 1.8; 1.003 in version 1.7); now 1.003 (0.905–1.106, version 1.9)
Inclusive purity, final cut	58.5/57.4/57.9%	57.3/56.6/56.9% without the Bragg-pion requirement (version 1.7)
Inclusive selection efficiency	17.5–17.6%	18.3–18.4% (version 1.7)
Largest alternative-model bias	1.18σ (inclusive); $-1.14/-1.16\sigma$ (proton-tagged $\delta p_T/p_n$)	1.10σ; $-0.99/-0.97\sigma$ (version 1.7); 1.21σ; $-1.08/-1.04\sigma$ (version 1.8); 1.21σ; $-1.19/-1.02\sigma$ and $+1.04\sigma$ (lower W_{had}) (version 1.9)
MCS term, FHC p_μ	8.2% bin-averaged, 0.56σ in the open bin	9.7%, 0.60σ (version 1.7); 10.1%, 0.65σ (version 1.8); replaced by the calibration term, 6.5%, 0.28σ (version 1.9)
RHC $W_{\pi p}$ cross section	-0.194	sign was a POT-convention artefact
Combined detector systematic	42.3%	24.6% at the fix; 17.6% (p_μ) in version 1.8, with the fractional treatment; 18.5% in version 1.9
Combined result / expected	0.910	1.027 at the fix (2026-08-28 pass)
FHC θ_μ A_C conditioning	0.015	now 0.51
Mean δp_T	0.742 GeV/c	now 0.264 GeV/c (beam frame)
Proton-tagged `detVar`	`n/a`, then 10.5–64.4%, then 10.5–108.5% (2026-09-06)	9.9–74.5% (median 30.9%, 2026-09-26 release); 6.8–47.5% (median 19.2%, version 1.8); now 5.5–55.0% (median 18.2%, version 1.9)
Proton-tagged σ, all obs.	2026-08-18 values	superseded 2026-08-31, and again by the re-thrown release of 2026-09-04 (current values in the note)
Cut-and-count totals	1.06/0.98/1.02; 1.03 ± 0.43 / 0.93 ± 0.42 / 0.98 ± 0.40 (borrowed systematic)	now the one-bin extraction 1.04 ± 0.37 / 0.92 ± 0.32 / 0.97 ± 0.34 (version 1.8)
Proton-tagged purity	47–48%	now 52/50/51%
Signal-region dirt (final cut)	1.1/1.0/2.1 (FHC/RHC/-comb.)	now 6.6/9.5/16.0; inclusive purity 58.8/57.7/58.2% now 58.5/57.4/57.9% (2026-10-01)
Control-region dirt (FHC)	49/0.1/10/1.9 (CC0π/multi-π/π^0/cosmic)	now 342/0.5/93/18.5 (2026-10-01)
Control-region global χ^2	pseudo-data 0.66/8; beam-on 8.11/8	now 2.29/8 and 6.03/8 (2026-10-01)
Ensemble offset and pull widths	2.5% at 2.1σ; widths 1.04/0.96	now 0.5–2.7% at $\leq 1.9\sigma$; widths 1.08/0.97/1.02; 2026-09-20: six ensembles (FHC/RHC/COMB, 46–100 members) 0.4–1.3% at $\leq 1.5\sigma$, widths 0.90–1.10; 2026-09-24: 100 members each, race-corrupted members re-run, 0.4–1.1% at $\leq 2.0\sigma$, widths 0.92–1.07; 2026-09-27 (binnings of the 0.50 criterion): widths 0.95–1.07 inclusive, 0.94–1.08 proton-tagged, offsets 0.3–1.1% at $\leq 2.0\sigma$; 2026-10-05 (version 1.7): widths 0.92–1.03 inclusive, 0.93–1.10 proton-tagged, one-dimensional offsets within 0.7% at $\leq 1.5\sigma$; 2026-10-09 (version 1.8, unfolded again): widths 0.92–1.04 inclusive, 0.92–1.09 proton-tagged, one-dimensional offsets within 0.9% at $\leq 1.5\sigma$; 2026-10-10 (version 1.9, five ensembles rebuilt): widths 0.92–1.04 inclusive, 0.89–1.04 proton-tagged, one-dimensional offsets within 1.4% at $\leq 1.5\sigma$
FHC beam-off in the signal region	128, then 13, then 30	now 23 (per-run gates, corrected Run-1 exposure)
Combined generator weights (FHC:RHC)	0.458:0.542	now 0.425:0.575 (2026-09-24)